Cooperation, Commitments, and the Value of the World Trading System

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Abstract

This paper asks how changes in the trading environment affect the welfare value of the world trading system. It extends a revealed-preference framework to represent the system as two restraints, cooperation in governments' objectives and commitments that cap tariffs at WTO bindings and preferential ceilings. In the calibrated 2017 tradable economy, positive tariffs below these ceilings imply that governments count a dollar of foreign real income as about half a dollar at home. Together, the two restraints raise world real income by about 0.81 percent relative to noncooperation. The 2018 to 2019 tariff conflict between the United States and China lowers world real income but raises the aggregate value of the remaining system by 9.5 percent. At the margin, the conflict makes raising tariffs on third-country imports less costly for both countries. The system preserves most of third countries' gains from diverted trade, mainly through commitments that limit tariff adjustment. Projected climate change instead leaves the system's aggregate value nearly unchanged, since large agricultural productivity changes affect a small share of modeled trade and manufacturing changes are small. The system therefore provides little additional mitigation of the projected income losses.

JEL classification: F13, F14, F51, F53, Q17, Q54
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1 Introduction

How do changes in trading environments affect the welfare value of the world trading system? Acting alone, a government with market power can shift part of a tariff's cost to foreign suppliers, creating an incentive to set tariffs above globally efficient levels. Trade agreements can raise welfare by restraining these tariffs. Exogenous changes in trade barriers and productivity can alter the costs of tariff restrictions and governments' incentives to impose them. Existing agreements can therefore become more or less valuable even when their rules remain unchanged. This paper measures that value in a quantitative model of tariff choice and examines how it changes in two applications, the 2018–2019 United States–China tariff conflict and projected climate change.

This paper distinguishes two forms of tariff restraint in the world trading system, cooperation and tariff commitments. Cooperation is represented by a weight on foreign real income in a government's objective, capturing the extent to which its tariff choices account for costs imposed abroad. Tariff commitments cap the tariffs a government can choose through WTO bindings and preferential schedules. They can themselves arise from international cooperation. The distinction concerns restraint through government objectives and through limits on policy choices. Both can explain a low tariff, but they can imply different responses to a changing environment. Under cooperation, a feasible tariff increase is attractive when its marginal domestic gain exceeds the weighted marginal loss abroad. A ceiling can prevent that increase even when the government would prefer it. Treating a tariff at its ceiling as a free choice can attribute to cooperation the restraint supplied by the tariff commitment.

The analysis extends the revealed-preference framework of Adão et al. (2026) by adding WTO bindings and preferential schedules to governments' feasible sets. The model is a multi-country, multi-sector Ricardian economy in which governments account for general-equilibrium effects when choosing tariffs. Positive tariffs sufficiently below their ceilings supply the interior observations used to estimate preferences, since governments could have raised them and did not. The weight on foreign real income is estimated to fit the balance between marginal domestic gains and foreign losses at these observed tariffs. In the 2017 calibration, a weight imposed as common to all governments is estimated at 0.50, so a dollar of foreign real income counts about half as much as a dollar at home. The model represents the world trading system as the regime in which governments keep the estimated cooperation and the existing tariff commitments. Its value is the real-income gain over a solved noncooperative equilibrium in which both are removed. Governments choose tariffs in each regime, taking other governments' choices as given, so the comparison uses solved policy equilibria rather than observed tariffs. Keeping one restraint alone shows what each protects separately.

The framework is applied to two changes whose magnitudes are informed by existing empirical research. The first draws on documented tariff changes from the 2018–2019 United States–China conflict. Bilateral tariffs are fixed across regimes at their preconflict and conflict rates, while governments choose the remaining eligible tariffs in each state. The comparison therefore

values the remaining system against a matched preconflict benchmark. The second applies end-of-century productivity projections under a high-emissions scenario, allowing all eligible tariffs to adjust. It asks how productivity changes across countries and sectors affect the income protected by existing restraint. The cases therefore examine how the system's value responds to changes in bilateral policy barriers and in production conditions.

In the baseline, the world trading system raises world real income by approximately 0.81 percent relative to noncooperation in the modeled tradable economy. Observed tariffs deliver 0.53 percent against the same benchmark, since the solved system selects lower tariffs than those observed, particularly in agriculture. At the estimated welfare weight, cooperation alone preserves 96 percent of the system's gain and tariff commitments alone 91 percent, because both prevent many of the same tariff increases. Their benefits are distributed differently. Cooperation weighs foreign income losses in levels and is relatively more valuable to several large economies, whose partners' tariffs impose large absolute losses. Tariff commitments protect market access even when the losses abroad are small in absolute terms and are relatively more valuable to many smaller economies.

The United States–China application shows that bilateral tariff increases between two large members can raise the value of restraint in other trade relationships. Higher bilateral tariffs pull the incentive to tax imports from third countries in two directions. They can weaken it, since lower tariffs on alternative suppliers can ease substitution away from imports made more expensive by the bilateral barriers. They can also strengthen it, because bilateral tariffs can divert purchases toward suppliers that are more costly before tax, and taxing those suppliers can partly reverse this distortion. In this calibration, the conflict lowers the marginal distortion cost of taxing other suppliers for both the United States and China, though that cost remains positive. Economies outside the pair initially gain in the aggregate from diverted trade, but unrestrained tariff adjustment across governments reverses those gains. The value of the remaining system rises by 9.5 percent in the world aggregate and also increases for every third country, although world real income falls and the two participants value it less. Commitments preserve most of those initial gains by limiting tariff adjustment, whereas cooperation alone moderates the adjustment without preventing the gains from becoming losses.

In the second application, projected climate change lowers world real income by about 2.3 percent both with and without the system, leaving its aggregate value nearly unchanged. In this model, a common proportional productivity decline across all countries and sectors leaves the system's value unchanged, so only relative productivity changes can alter it. These changes are larger in agriculture, which accounts for only 3.5 percent of modeled trade, and smaller in manufacturing, which accounts for most of it. The system's value to individual economies nevertheless moves in both directions, reflecting how productivity changes at home and abroad interact with each economy's trade patterns. Six reforms of the world trading system that tighten agricultural tariff commitments or strengthen cooperation do little to offset the world income loss.

The analysis relates to research on the rationale, enforcement, and value of trade agreements. Bagwell and Staiger (1999) develop the terms-of-trade rationale, while Bagwell and Staiger (1990) and Maggi (1999) study how repeated interaction can sustain cooperation. Amador and Bagwell (2013) and Beshkar et al. (2015) analyze tariff ceilings. Nicita et al. (2018) find tariff-setting patterns consistent with noncooperation when applied tariffs lie below their bindings. Quantitative studies evaluate the costs of tariff wars (Ossa, 2014; Lashkaripour, 2021) and the welfare implications of negotiation rules (Bagwell et al., 2021). Handley and Limão (2017) quantify welfare gains from reduced policy uncertainty, a channel outside the present static framework. The closest methodological reference is Adão et al. (2026), who infer foreign welfare weights from observed tariffs and interpret them as summarizing the effects of formal and informal institutions. Here, explicit ceilings separate tariff commitments from the cooperation estimated from interior tariffs, allowing both restraints to be valued separately and jointly as the trading environment changes.

The applications connect this institutional comparison to research on trade conflict and climate adaptation. Fajgelbaum et al. (2020) estimate the price, trade, and welfare effects of the United States tariff increases and foreign retaliation, Fajgelbaum et al. (2024) document export opportunities for countries outside the conflict, and Adão et al. (2025) use the episode to test quantitative welfare predictions. The climate application uses productivity projections assembled by Nath (2025) and relates to studies of adaptation through production, trade, and migration (Costinot et al., 2016; Gouel and Laborde, 2021; Cruz and Rossi-Hansberg, 2024). These studies take trade policy as given. Here, governments choose their remaining eligible tariffs under alternative institutions, and the analysis measures how each shock changes the income protected by the two restraints in the world trading system.

Section 2 presents the framework for tariff choice under cooperation and tariff commitments and defines the welfare value of these restraints. Section 3 describes the data, calibration, and estimation of foreign welfare weights. Section 4 reports the baseline value of the world trading system and its distribution across economies. Sections 5 and 6 examine how the United States–China tariff conflict and projected climate change affect that value. Section 7 assesses sensitivity and robustness, and Section 8 concludes.

2 Framework

The framework separates cooperation in governments' objectives from tariff commitments that restrict their tariff choices. It defines the welfare value of these restraints and distinguishes exposure to a changing trading environment from the effects of tariff adjustment.

2.1 Trade, cooperation, and tariff commitments

The trade environment follows Eaton and Kortum (2002), with multiple sectors as in Caliendo and Parro (2015) and no intermediate inputs. Countries are indexed by i, j and sectors by

s. Each country has a fixed labor endowment that can move freely across domestic sectors. Country-sector productivity follows a Fréchet distribution with scale T_{js} and shape θ_s. The index $a_{js} = T_{js}^{1/\theta_s}$ measures productivity in units that scale output per worker. Households have fixed Cobb–Douglas expenditure shares across sectors. Appendix A states the competitive equilibrium and its representation in relative changes.

Let $t_{ji}^s \geq 0$ be the tariff imposed by importer i on goods from exporter j in sector s. Each eligible bilateral-sector tariff is a separate policy instrument, indexed by ℓ when the exporter and sector need not be shown. Trade also incurs iceberg costs. Tariff revenue is rebated to households, and trade is balanced. Real income is $W_i(\mathbf{T}, \mathbf{t}) = I_i/P_i$, where nominal income I_i includes labor income and tariff revenue. Superscript 0 denotes the calibrated allocation. Multiplying real income by the fixed price index P_i^0 expresses it in common monetary units at baseline prices, allowing domestic and foreign income to enter the same government objective.

Government i chooses its tariffs to maximize

$$G_i(\mathbf{T}, \mathbf{t}; \boldsymbol{\lambda}) = P_i^0 W_i(\mathbf{T}, \mathbf{t}) + \sum_{j \neq i} \lambda_{ij} P_j^0 W_j(\mathbf{T}, \mathbf{t}), \qquad \lambda_{ij} \geq 0. \tag{1}$$

The weight λ_{ij} measures how government i values a dollar of real income in country j relative to a dollar at home, with both valued at baseline prices. Adão et al. (2026) interpret foreign welfare weights as summarizing the effects of formal and informal institutions. Here tariff commitments enter separately as constraints, and the weight is estimated from positive tariffs sufficiently below their ceilings. Cooperation therefore denotes the internalization of foreign income effects in government objectives, conditional on the modeled tariff commitments.[1] The vector $\boldsymbol{\lambda}$ collects these weights. The weights capture the internalization of foreign income effects and need not represent literal concern for foreign households. Zero weights imply domestic real-income maximization. A positive weight makes a tariff increase less attractive when it harms the corresponding partner. The weights are fixed and the objective is linear in foreign real income, so a partner's income loss does not increase the marginal weight on its income.

Tariff commitments restrict policy independently of these weights. Let $\mathcal{D}_i$ be the set of tariffs available without institutional ceilings, subject to nonnegativity and the numerical upper bounds specified in Section 3.1. A WTO binding or preferential schedule supplies a ceiling $\bar{t}_{i\ell}$ on each covered instrument. With tariff commitments, the feasible set is

$$\mathcal{D}_i^I = \{\mathbf{t}_i \in \mathcal{D}_i : t_{i\ell} \leq \bar{t}_{i\ell} \text{ for covered instruments } \ell\}. \tag{2}$$

A positive tariff below its ceiling can reflect a choice to limit harm abroad. A tariff at its ceiling can instead reflect a constraint on a desired increase. When the ceiling equals the lower bound,

[1] Self-enforcing cooperation provides one possible foundation for this behavior. Bagwell and Staiger (1990) and Maggi (1999) study cooperation in repeated tariff games, including the role of multilateral institutions in sustaining negotiated commitments, and Mei (2020) develops a quantitative analysis. The present framework does not identify the enforcement mechanism sustaining the estimated behavior. The estimate rises when ceiling-class cells enter the sample or when preferential ceilings are removed from the domain (Sections 3.3 and 7), which is consistent with the weight absorbing restraint that constraints would otherwise represent.

Table 1: Policy regimes for the remaining tariff choices

Regime	Foreign welfare weights	Institutional ceilings
Full system (sys)	Estimated	Retained
Tariff commitments only (com)	Zero	Retained
Cooperation only (coop)	Estimated	Removed
Noncooperation (nc)	Zero	Removed

Notes: Institutional ceilings apply to the remaining tariff choices, while imposed tariffs may exceed them. Fixed background policies remain unchanged. Tariffs remain nonnegative and subject to numerical upper bounds in all regimes. The estimated welfare weights are retained in full, including the United States–China weights, in sys and coop.

as with a preferential ceiling of zero, the tariff is fixed and supplies no first-order restriction on preferences. These distinctions determine which observed tariffs can be used as interior choices when estimating cooperation.

2.2 Policy regimes and their value

Each application compares an initial trading environment, indexed by $r = 0$, with a changed environment, indexed by $r = 1$. The climate application changes productivity. The conflict application changes bilateral tariffs specified by the experiment, referred to as imposed tariffs. Within each environment, these tariffs are held fixed at the same rates in all four policy regimes.

The four regimes in Table 1 retain both restraints, one restraint, or neither. In each regime, governments choose their remaining eligible tariffs within the applicable bounds, taking the other governments' tariffs as given. Their choices account for general-equilibrium effects on wages, expenditures, and tariff revenue. A policy equilibrium is a tariff profile in which each government's choices maximize its objective given the others' choices. Denote this profile by $\mathbf{x}^{R,r}$ for regime R in environment r.

Let $W_i^R(r)$ be country i's real income at the policy equilibrium under regime R in environment r. The value of that regime is its log income gain relative to noncooperation in the same environment,

$$v_i^R(r) = \log W_i^R(r) - \log W_i^{\mathrm{nc}}(r), \qquad R \in \{\mathrm{sys}, \mathrm{com}, \mathrm{coop}\}. \tag{3}$$

The full-system regime represents the world trading system as modeled here, with cooperation and tariff commitments retained together. Institutional value refers to its value $v_i^{\mathrm{sys}}(r)$ unless another regime is named. In the conflict application, it measures the value of the remaining restraints conditional on the imposed bilateral tariffs.

The values under tariff commitments alone and cooperation alone measure each restraint in the absence of the other. They need not sum to the value of the system because the two mechanisms can prevent the same tariff increases. Nor must every national value be positive, since a regime changes both a country's own policies and those of its partners.

Aggregate value averages countries' log real-income gains relative to noncooperation, using each country's share of baseline income within the reporting group as its weight. The weights remain fixed across environments and regimes. Results are reported in log percentage points

and describe the calibrated tradable economy rather than world GDP. A percentage change in institutional value expresses the change relative to its initial value. Appendix B.2 gives the formal aggregation rule and reporting groups.

2.3 Exposure and policy adjustment

The main decomposition first changes the environment while holding each regime's remaining tariffs at their initial values. This exposure step allows trade and prices to respond in competitive equilibrium. It then evaluates tariff adjustment in the changed environment. This sequence measures whether policy responses amplify or mitigate the income effect of the shock relative to retaining initial tariffs. It defines counterfactual comparisons within a static model, rather than a predicted timeline of adjustment.

Let $u_i(r, \mathbf{x})$ denote country i's log real income in environment r when the remaining tariffs are $\mathbf{x}$. Using the equilibrium tariffs $\mathbf{x}^{R,r}$ defined above, exposure and adjustment are

$$\begin{aligned} E_i^R &= u_i(1, \mathbf{x}^{R,0}) - u_i(0, \mathbf{x}^{R,0}), \\ A_i^R &= u_i(1, \mathbf{x}^{R,1}) - u_i(1, \mathbf{x}^{R,0}). \end{aligned} \tag{4}$$

They give the exact decomposition

$$\Delta v_i^R = (E_i^R - E_i^{\mathrm{nc}}) + (A_i^R - A_i^{\mathrm{nc}}). \tag{5}$$

Thus, exposure raises institutional value when the initial policies under restraint protect income better against the change in the environment. Adjustment raises value when the subsequent tariff response is less costly under restraint. Neither component has a general sign in the multi-country model. Appendix B.3 gives the local counterpart and a further split between own and partner responses.

Productivity homogeneity provides a useful benchmark. If a_{js} rises by the same factor in every country and sector, real income rises by that factor at every tariff profile. Each government objective is scaled equally, leaving best responses unchanged. Corresponding policy equilibria therefore have the same log income gaps. Corollary 1 in Appendix A.4 establishes this result under the stated domains. A large common productivity loss can leave institutional value unchanged. Relative productivity changes, however, alter trade patterns and tariff incentives and must be evaluated quantitatively.

3 Data and Estimation

Tariff ceilings determine both the policy choices available in counterfactuals and the observations used to estimate cooperation. This section constructs the calibrated economy and these ceilings, then estimates foreign welfare weights from positive tariffs sufficiently below them.

3.1 Calibration and tariff domains

The calibration uses 2017 trade and production for 34 named economies and a rest-of-world aggregate, with the European Union treated as one economy. The three sectors are agriculture, extraction, and manufacturing. Bilateral product trade comes from BACI, applied tariffs from WITS, and production from FAOSTAT and UNIDO, supplemented by OECD data for specified gaps. The sectoral trade elasticities are 8.11, 15.72, and 5.62. They average industry values based on Caliendo and Parro (2015) with equal weights across ISIC two-digit divisions, incorporating three industry reassignments documented in Appendix C.1. The appendix also documents the sources and concordances.

Trade and production are reconciled to obtain positive domestic sales. Recorded trade deficits are then removed at unchanged technology and observed tariffs using exact hat algebra (Dekle et al., 2008). The balanced allocation supplies the baseline trade shares, income weights, and price indices. Production data measure gross output, whereas the model uses labor alone. The resulting welfare measure therefore describes the calibrated tradable economy, rather than national-account GDP or consumption.

There are 32 optimizing governments and 2,989 bilateral-sector tariff cells. Saudi Arabia, the United Arab Emirates, and the rest of world retain their observed tariffs while participating in the competitive equilibrium. The first two lack consolidated WTO schedules in the assembled data. Iran optimizes without WTO ceilings and is excluded from estimation.

Institutional ceilings combine WTO bindings and preferential schedules. WTO ceilings use verified end-of-2017 rates where available and documented proxies otherwise. Named preferential relationships receive their observed tariffs as ceilings. Rates are aggregated with fixed baseline import weights. Products with unknown bindings receive the cell's import-weighted average of known capped rates, or the cap if none is known. They account for 10.3 percent of active imports before the deficit adjustment. Ceilings are raised to observed tariffs where needed for feasibility, affecting 132 cells that cover 15.1 percent of the same import base. The numerical upper bound is the larger of 200 percent and the observed tariff. Appendix C.3 documents these rules and the treatment of preferential components of rest-of-world trade.

Cells are classified by their feasible tariff range. Singleton cells have no feasible tariff change. Among the remaining cells, zero cells have a zero tariff that can be raised. Interior cells have a positive tariff at least five percentage points below their ceiling. Ceiling cells have a positive tariff at the ceiling or less than five points below it.

The estimating sample contains 959 interior cells from 27 governments, covering 13.5 percent of active imports in the balanced baseline. Named preferential cells are excluded from this sample because their ceilings equal observed tariffs. The five-point margin accommodates differences arising from concordances and aggregation and is varied in sensitivity checks. An interior sectoral cell can contain products at zero or at their individual bindings, so product-level interior coverage supplies an additional diagnostic.

3.2 Revealed-preference estimation

For an interior observed tariff $t_{i\ell}$, the first-order condition of (1), evaluated at the calibrated allocation, is

$$I_i^0 \frac{d \log W_i}{dt_{i\ell}} + \sum_{j \neq i} \lambda_{ij} I_j^0 \frac{d \log W_j}{dt_{i\ell}} = 0. \tag{6}$$

The general-equilibrium derivatives are computed as in Appendix A.3. Define $y_{i\ell} = d \log W_i / dt_{i\ell}$ and $X_{i\ell} = -\sum_{j \neq i} (I_j^0 / I_i^0) d \log W_j / dt_{i\ell}$. Imposing the common weight $\lambda_{ij} = \lambda$ gives

$$y_{i\ell} = \lambda X_{i\ell} + \varepsilon_{i\ell}, \qquad \ell \in \mathcal{E}_i, \tag{7}$$

where $\mathcal{E}_i$ denotes the interior sample. Least squares without an intercept estimates the weight from the relation between marginal domestic gains and foreign losses across cells, both normalized by domestic income. These margins vary with trade shares, observed tariffs, and trade elasticities. A common weight is a parsimonious restriction given only three sectoral observations for each bilateral relationship. This restriction also applies to preferential partners and does not allow their foreign welfare weights to differ systematically from those of other partners.

Interpreting the slope as cooperation requires $\mathbb{E}[X_{i\ell} \varepsilon_{i\ell} \mid \ell \in \mathcal{E}_i] = 0$. The residual can contain omitted policy motives, measurement error, aggregation effects, and departures from the common-weight restriction. Evaluating welfare margins at observed tariffs introduces a simultaneity concern. Selection into the interior sample can also correlate foreign margins with omitted motives. An instrumental-variable check uses the foreign margin evaluated at free trade. This removes its direct dependence on observed tariffs, but retains an exclusion restriction and does not resolve selection.

3.3 Estimates and model fit

The baseline foreign welfare weight is 0.50, with an importer-clustered standard error of 0.07. Allowing dependence between cells that share a country in either role raises the standard error to 0.16. The pooled estimate is applied to all 32 optimizing governments, including those without estimation observations. A dollar of foreign real income receives approximately half the weight of a domestic dollar in the estimated objective. The instrumental-variable estimate is 0.41, with a standard error of 0.08 and a first-stage F statistic of 153.9. Table 2 reports the estimates, and Figure 1 shows the underlying margins.

Extending the equality regression to ceiling and zero cells raises the coefficient. Using all 2,502 cells in the broad sample gives 0.85. Adding only zero cells gives 0.55, while adding nonpreferential ceiling cells gives 0.58. Preferential ceiling cells account for much of the remaining increase. These comparisons illustrate the importance of distinguishing interior observations from cells at or near tariff bounds. They also change sample composition, so the differences cannot be attributed entirely to a causal effect of constraints on inferred cooperation.

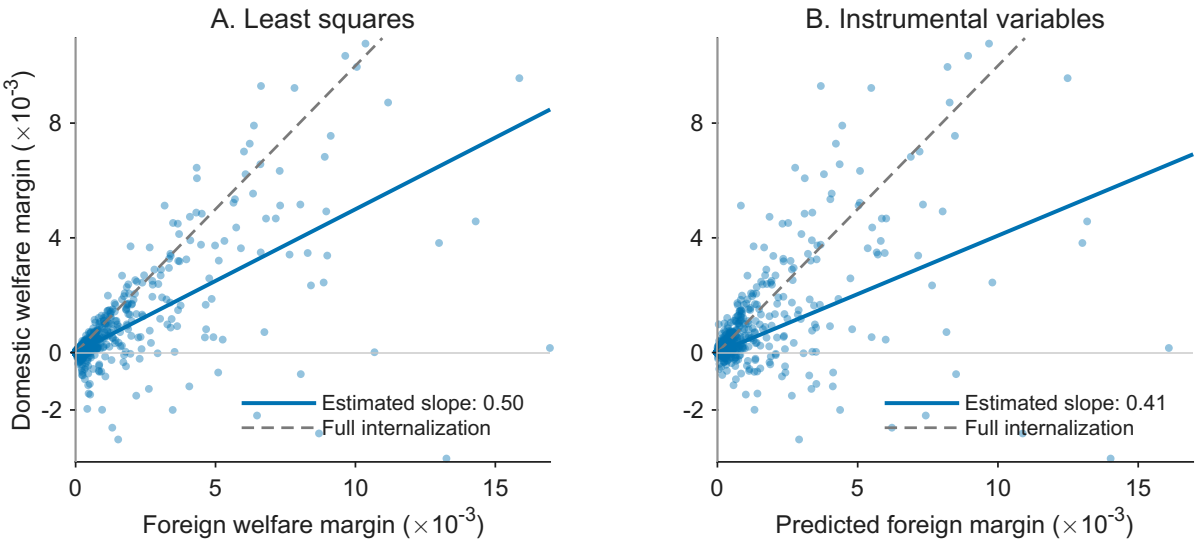
Product-level coverage matters within the interior sample. Requiring at least half of a

Table 2: Estimates of the foreign welfare weight

	$\widehat{\lambda}$	s.e.	Cells
Main estimates and broader-sample comparison			
Baseline, interior cells	0.50	0.07	959
Free-trade instrument	0.41	0.08	959
Interior plus ceiling and zero cells	0.85	0.05	2,502
Additional cells, interior coverage, partner groups, and fixed effects			
Interior plus zero cells	0.55	0.05	1,071
Interior plus nonpreferential ceiling cells	0.58	0.05	1,450
Interior coverage of at least one quarter	0.27	0.08	627
Interior coverage of at least one half	0.21	0.10	488
Weighted by interior coverage	0.31	0.07	959
Advanced partners	0.57	0.07	959
Developing partners	0.47	0.07	959
Importer fixed effects added	0.48	0.07	959

Notes: The estimated welfare weights and their standard errors are dimensionless. The Cells column reports the number of tariff observations. Both welfare margins are divided by the importing government's baseline income. Regressions have no intercept except in the fixed-effects specification. Standard errors are clustered by importer. All rows use least squares except the row labeled "Free-trade instrument," which instruments the foreign margin with its value at free trade. The broad sample adds 1,431 ceiling and 112 zero cells to the interior sample and excludes 398 singletons and Iran's 89 interior cells. The first two rows in the lower panel add zero cells or the 491 ceiling cells of nonpreferential partners as equalities. Interior coverage follows Appendix C.3. The partner rows are two slopes from one regression. Appendix Table G5 reports further specifications.

Figure 1: Domestic gains and foreign losses from marginal tariff increases



Notes: Each point is an interior government–partner–sector cell. The vertical axis is $y_{i\ell}$ in (7). Panel A uses $X_{i\ell}$. Panel B uses its no-intercept first-stage prediction from the free-trade instrument. Both margins are multiplied by 1,000, with common axis limits across panels. Solid lines are the full-sample least-squares and instrumental-variable fits. Dashed lines show full internalization, $y = X$. Horizontal limits cover the union of the two first–ninety-ninth percentile ranges, and vertical limits use the corresponding percentiles of y. These display limits omit 32 and 30 observations, respectively. All 959 cells enter both estimates.

Table 3: The value of the world trading system in 2017

	World	Median
Solved policy equilibria		
Tariff commitments only	0.73	1.78
Cooperation only	0.77	1.30
Full system	0.80	2.04
Reference schedules		
Observed 2017 tariffs	0.53	1.62
Free trade	0.84	2.01

Notes: Entries are real-income gains relative to the noncooperative equilibrium at baseline productivity, calculated as 100 times log income ratios and reported in log percentage points. World values use fixed baseline-income weights over the 34 named economies, and medians cover the 32 optimizing governments. The first panel reports the regimes defined in Table 1. The second panel values two schedules that are not equilibria of the policy game. Appendix Table D1 reports national values by regime.

cell's imports to lie on interior products lowers the estimate to 0.21, and weighting by interior coverage gives 0.31. These changes can reflect both constraints within aggregated cells and the different partners and sectors retained. Appendix C.5 reports coverage and inference diagnostics. Counterfactuals at lower weights assess the economic importance of this sensitivity.

The common objective fits observed ceiling contacts better than observed zero tariffs. Its necessary optimality condition holds on approximately 91 percent of imports at exact observed ceiling contacts, excluding singletons, but only 2.2 percent of imports in zero cells. Zero cells cover 0.43 percent of active imports and are concentrated in extraction. Input-cost motives for low tariffs are absent from a model without intermediates. The estimated objective does not rationalize the entire observed schedule, so the analysis reports its solved equilibrium separately from the welfare delivered by observed tariffs.

4 The Value of the World Trading System in 2017

This section evaluates cooperation and tariff commitments in the calibrated 2017 economy. It reports aggregate and national gains and examines unilateral departure.

4.1 Aggregate gains and overlapping restraint

The world trading system raises world real income by 0.803 log percentage points, about 0.81 percent, relative to noncooperation. This amounts to about 95 percent of the gain from free trade, which is 0.84 points against the same benchmark. Tariff commitments alone deliver 0.73 points and cooperation alone 0.77 points, or 91 and 96 percent of the value of the system. The median gain among the 32 optimizing governments is 2.04 points. Several large economies have smaller proportional gains and receive large income weights, helping explain why the world value is below the median. Table 3 summarizes these comparisons.

Observed 2017 tariffs deliver 0.53 log percentage points against the same noncooperative benchmark. This comparison does not use the estimated foreign welfare weight. The solved system delivers a larger gain and selects lower tariffs on average, particularly in agriculture. In

advanced economies, solved agricultural tariffs average 2.1 percent, compared with 12.7 percent in the observed schedule (Appendix Table D2). The applications use solved equilibria so that government objectives and policy rules remain fixed within each regime while eligible tariffs can respond to the changing environment.

The large stand-alone values reflect overlapping restraint. Both mechanisms prevent many of the same high tariffs. For example, tariff commitments alone reduce advanced-economy manufacturing tariffs from about 16 percent to 2.6 percent, and the system reduces them further to 1.7 percent. A local approximation in which efficiency losses are quadratic in tariff wedges illustrates why further reductions yield smaller gains once tariffs are low. This helps explain why adding a second restraint produces a smaller aggregate gain. The quantitative comparisons also include terms-of-trade effects and interactions across markets.

Cooperation also affects whether governments reach their ceilings. Under tariff commitments alone, tariffs at their ceilings cover 82.4 percent of baseline imports in cells where tariffs can vary. Adding cooperation lowers this share to 69.3 percent. Appendix D.2 reports the contact counts and the transitions between regimes.

4.2 National gains and unilateral departure

Every optimizing government gains from the system, but proportional gains vary substantially. They range from 0.13 log percentage points for China to 6.37 for Mexico. Norway and Switzerland gain more than five points, while the United States gains 0.55. Appendix Figure D1 compares national values under the solved system and observed tariffs. Under observed tariffs, the comparison with noncooperation is negative for Argentina, India, Korea, and Turkey. Large domestic markets generally reduce dependence on foreign market access, and governments with market power can offset some foreign protection through their own terms-of-trade gains. Smaller economies are generally more exposed to barriers imposed by major partners.

The two restraints distribute these gains differently. Cooperation alone is more valuable than tariff commitments alone for six governments, the United States, China, Japan, India, Argentina, and the European Union. Tariff commitments alone are more valuable for the other 26 (Appendix Table D1). The estimated weight is common to all partners and applies to foreign income losses in levels. Holding the domestic benefit fixed, a larger absolute loss abroad therefore makes a tariff increase less attractive. This mechanism can make cooperation especially valuable to large exporting partners. A ceiling protects a smaller exporter's market access even when the associated foreign-income loss is small in absolute terms.

Adding tariff commitments to cooperation can lower a country's income. For all six governments above, the system is worth less than cooperation alone. The United States gains 0.55 points under the system and 0.94 under cooperation alone. Ceilings constrain a government's own tariff choices while protecting its access to foreign markets, and the net effect need not be positive. These comparisons measure the combined effect of changing ceilings at home and abroad, without separating their contributions.

The gains from mutual restraint coexist with domestic-income gains from unilateral departure. Every optimizing government can raise its domestic real income by setting its foreign welfare weights to zero and choosing tariffs without its institutional ceilings, while partners retain their system tariffs. A United States departure raises its income by 0.899 log percentage points. All 33 other named economies lose, with an income-weighted aggregate loss of 0.273 points (Appendix D.3). The conflict exercise in Section 5 instead allows the remaining eligible tariffs to adjust, making the response under each regime central to the comparison.

5 The United States–China Conflict

Bilateral tariff increases can change the value of agreements governing other trade relationships by redirecting trade and altering incentives for further protection. The 2018–2019 United States–China conflict provides a documented episode involving two large markets. Fajgelbaum et al. (2024) document export opportunities for countries outside the conflict, but those opportunities do not establish how much income the remaining system protects against subsequent tariff responses. This section examines how the conflict changes that value for the United States, China, and third countries, and how the contributions of cooperation and tariff commitments differ.

The experiment takes the bilateral tariff schedules as given and applies them to the 2017 economy. The six bilateral-sector tariffs are fixed in every regime, first at their 2017 rates and then at conflict rates ranging from approximately 6 to 34 percent, up from single digits. Governments choose all remaining eligible tariffs, while productivity, nontariff trade costs, and labor endowments remain fixed.[2] Appendix E reports the imposed schedules and their construction.[3]

In each state, the value of the remaining system is its real-income gain over noncooperation conditional on the same imposed bilateral tariffs. The comparison uses a matched preconflict benchmark in which those tariffs are also fixed, rather than the unrestricted baseline in Section 4.

5.1 Income effects and the value of the remaining system

The conflict lowers world real income but raises the value of the remaining restraints. Income falls by 0.054 log percentage points with the remaining restraints, compared with 0.118 without them. The difference in losses raises institutional value from 0.672 to 0.735 points, an increase of 9.5 percent (Table 4).

Third countries drive the increase. Their aggregate value rises from 1.185 to 1.330 log percentage points, or 12.2 percent, since they gain 0.016 points of income with the remaining

[2]Regimes with cooperation retain all estimated welfare weights, including those between the United States and China. Section 7 considers setting these two weights to zero.

[3]The covered products account for 99.39 percent of raw bilateral imports into the United States and 94.89 percent into China. Ordinary rates come from annual schedules, so the indices proxy the end-2019 schedule rather than reproduce it exactly. Appendix E documents the national-line allocation, the treatment of missing rates and exclusions, and the alternative inputs used in Section 7.

Table 4: Conflict and the value of remaining restraint

	Institutional value			Income change	
Population	Preconflict	Conflict	Change	System	Noncoop.
World	0.6718	0.7354	0.0636	-0.0543	-0.1179
Third countries	1.1851	1.3300	0.1449	0.0163	-0.1286
United States	0.3736	0.2914	-0.0822	-0.1685	-0.0863
China	0.0271	0.0238	-0.0033	-0.1168	-0.1135

Notes: All entries are in log percentage points, with income comparisons equal to 100 times differences in log real income. Preconflict and Conflict measure the full system's income gain over noncooperation at the same imposed bilateral schedule, and Change is their difference. The Income change columns compare conflict with preconflict income within each regime. Their difference equals Change before rounding. Aggregates use fixed baseline-income weights over 34 named economies for the world and 32 for third countries. Results describe the calibrated tradable economy, whose production inputs measure gross output, rather than the episode's realized GDP effects.

Table 5: Third-country income effects before and after remaining tariffs adjust

Regime	Exposure	Adjustment	Total
Full system	0.0166	-0.0003	0.0163
Tariff commitments only	0.0170	-0.0006	0.0164
Cooperation only	0.0134	-0.0747	-0.0613
Noncooperation	0.0132	-0.1419	-0.1286

Notes: Entries are changes in log real income multiplied by 100, reported in log percentage points and averaged using fixed baseline-income weights over the 32 named third countries. Total compares the conflict equilibrium with the matched preconflict equilibrium under the same regime. Exposure imposes the conflict bilateral tariffs while holding remaining tariffs at that regime's preconflict rates. Adjustment measures the additional income change when remaining tariffs are reoptimized. Components sum to the total before rounding and follow equation (4). Subtracting the noncooperative row gives changes in institutional value and their components. Appendix Table E4 reports the corresponding world, United States, and China results.

restraints but lose 0.129 without them. The values for the United States and China instead fall by 0.082 and 0.003 points. Both lose income in either case, but both still value the remaining restraints positively.

5.2 Why third countries value the remaining restraints more

The initial reallocation of trade gives third countries similar aggregate income gains with and without the remaining restraints. This initial effect, the exposure component of Section 2.3, therefore contributes little to the increase in institutional value. The main difference arises when governments adjust their other tariffs (Table 5).

Without the remaining restraints, tariff adjustment lowers third-country income by about 0.14 log percentage points and reverses the initial gain. With them, the additional loss is negligible. This difference accounts for about 98 percent of the increase in institutional value. Initial access to diverted trade is less valuable when trading partners remain free to raise barriers against the countries receiving it. Almost all of the adjustment difference comes from partners' tariff responses, including the indirect effects of their tariffs on other exporters (Appendix Table E9, Panel A).

Both conflict governments prefer higher tariffs on other suppliers after the conflict. Holding other governments' remaining tariffs at their preconflict noncooperative rates, the United States

best-response mean rises by approximately 4.4 percentage points and China's by 2.0. One reason is that buyers compare prices including tariffs, but tariff revenue remains in the importing country. Bilateral barriers can therefore divert purchases toward suppliers that are cheaper to buyers but more costly before tariffs. Taxing those alternatives can partly reverse this substitution, reducing the additional distortion from further protection. A local comparison of the same small tariff adjustment before and after the conflict finds lower, but still positive, marginal distortion costs for both governments in this calibration. A larger terms-of-trade gain also strengthens the United States incentive (Appendix E.4). These calculations explain their initial incentives without attributing the entire third-country reversal to their policies.

Tariff commitments limit these tariff increases more strongly in this experiment. A binding ceiling prevents a desired increase. Cooperation permits an increase when its domestic gain exceeds the weighted foreign loss. With ceilings retained, both governments' mean tariffs on other suppliers change by less than 0.1 percentage point. Under cooperation alone, both means rise, but by less than under noncooperation. The conflict-induced increase in third-country value is approximately the same under tariff commitments alone and the system, but less than half as large under cooperation alone (Appendix Table E5).

5.3 National outcomes among third countries

Every named third country values the remaining system more after the conflict, whether or not it gains income. In the principal solution, 22 third countries gain income with the remaining restraints and ten lose, while all 32 lose without them.[4] Figure 2 compares preconflict and conflict values across optimizing governments, and Appendix Table E3 reports all named economies. A country can thus become more dependent on existing restraint even when diverted trade does not compensate it for the conflict's other effects.

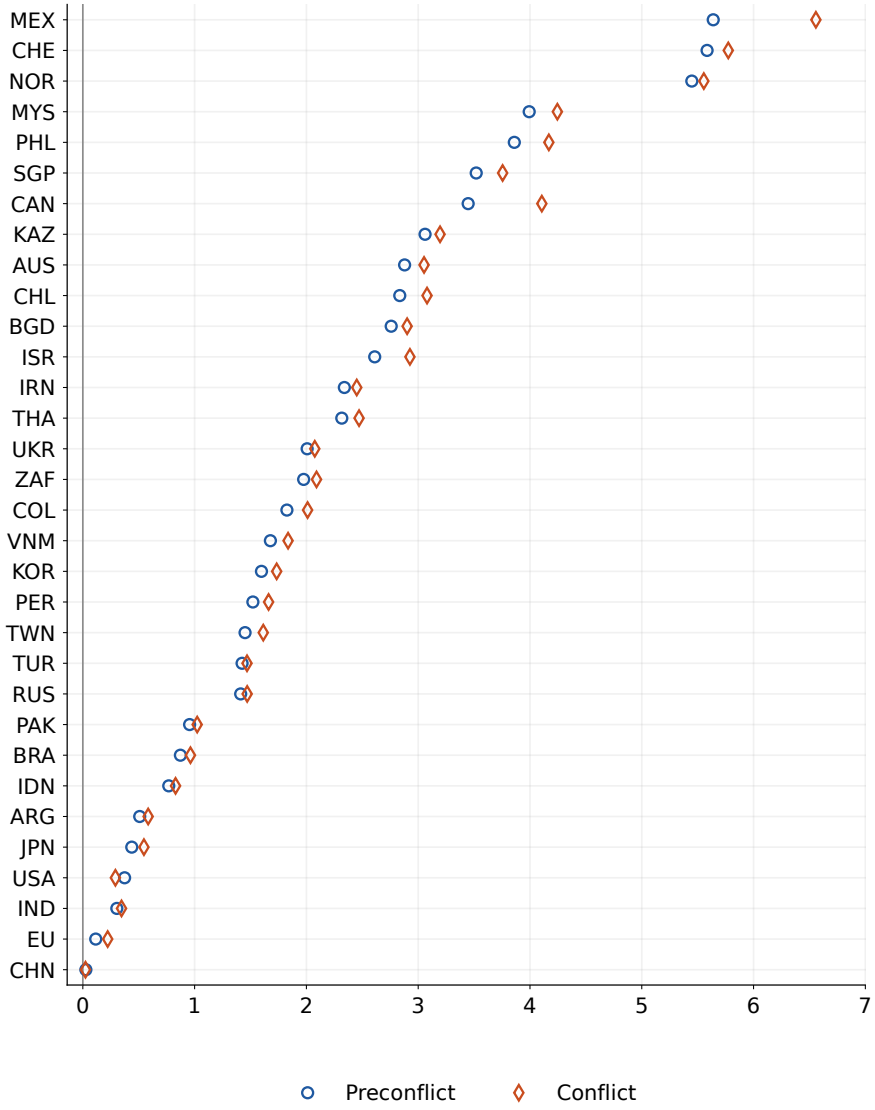
Mexico and Canada have the largest increases in value measured in log percentage points, corresponding to about 16 and 19 percent of their respective preconflict values. Mexico gains approximately 0.2 log percentage points of income from the conflict with the remaining restraints and loses 0.7 without them. Canada shows the same pattern. These outcomes are consistent with their dependence on United States market access, which the remaining restraints help preserve when the conflict raises incentives for further protection.

5.4 Outcomes for the United States and China

The value of the remaining system falls by 0.08 log percentage points for the United States. Most of the decline arises before remaining tariffs adjust. With restraint, low tariffs on other suppliers make the bilateral increases strongly discriminate against Chinese goods. Without restraint, the noncooperative tariffs on other suppliers are high even before the conflict, so the increases on Chinese goods can partly offset the initial tariff advantage of Chinese goods. This

[4] Brazil's near-zero income change switches sign across the two accepted starting profiles, which reduces the number of income gains with the remaining restraints to 21 in the alternative solution.

Figure 2: The value of the remaining trading system under preconflict and conflict tariffs



Notes: Circles and diamonds report values at the preconflict and conflict bilateral tariffs, respectively. Each value is the full system's real-income gain over noncooperation under the same imposed bilateral schedule, measured as 100 times the log income ratio and reported in log percentage points. The 32 optimizing governments are ordered by decreasing preconflict value. Saudi Arabia and the United Arab Emirates, which enter the third-country aggregate but retain fixed tariffs, are excluded.

difference in sourcing incentives helps explain why the imposed barriers are more costly with the remaining restraints. Subsequent adjustment reinforces the decline in the main decomposition. As Section 5.2 explains, the conflict strengthens the incentive to tax other suppliers, and restraint limits the domestic gains from that response, which partners' tariff responses only partly offset. Under cooperation alone the United States instead gains income, partly because partners that weigh its income cut their tariffs on its goods.

China experiences a smaller decline, 0.003 log percentage points. Before remaining tariffs adjust, the system cushions the conflict's income effect. Its advantage under the United States tariff increases outweighs its disadvantage under China's own increases. Access to alternative

export markets could contribute to this cushioning, although the experiments do not distinguish export redirection from other general-equilibrium effects. Subsequent tariff adjustment removes the initial advantage, mainly through partners' policies in the reported decomposition. China's small net change therefore reflects offsetting effects of the imposed barriers and the subsequent policy responses, rather than an absence of either effect. Appendix E reports the initial tariffs, the decompositions, and the cooperation-only results behind both paragraphs.

Together with the third-country results, these findings show that a bilateral conflict can strengthen the welfare case for maintaining agreements elsewhere while reducing their value to the participants. The case is collective rather than individual. Third countries as a group are better off keeping the remaining restraints. A single government might still gain by departing while its partners stay restrained, as every optimizing government can at the baseline (Section 4.2). Whether the threat of retaliation deters such departures is an enforcement question outside the static model.

6 Climate Change

Exogenous changes in comparative advantage alter what tariff restraint protects, because they change which goods countries trade, with whom, and how much each government gains from restricting that trade. Climate change is the leading such change in prospect. Its projected productivity losses are concentrated in agriculture, the most protected sector, and differ widely across countries, so they shift comparative advantage rather than scale the world economy down uniformly. This section asks whether existing restraints protect more or less income after these changes and whether reforms of the restraints would add mitigation. The aggregate value changes little, but national outcomes differ because productivity changes abroad matter as well as those at home.

6.1 Income effects and the value of the system

The experiment applies the projections assembled by Nath (2025) for 2080–2099 under RCP 8.5 to the 2017 economy. Agriculture and manufacturing productivity change, while extraction productivity and all other primitives remain fixed. Governments choose all eligible tariffs in both productivity states. Appendix F.1 documents the inputs and proxy assignments. The exercise evaluates these productivity changes rather than forecasts the end-of-century world economy.

Projected climate change lowers world real income by approximately 2.32 log percentage points both with and without the system (Table 6). Institutional value falls by only 0.0034 points, or 0.4 percent of its baseline value of 0.803. The system retains nearly the same proportional income benefit, but does not mitigate the projected aggregate loss, which is marginally larger with the restraints.

A common proportional productivity decline lowers income equally across regimes and leaves their log income gap unchanged, as Corollary 1 in Appendix A.4 establishes. Using baseline

Table 6: Climate change, world income, and the value of the system

	World income change		
Productivity changes applied	With the system	Without it	Change in value
Both sectors (main experiment)	−2.323	−2.320	−0.0034
Agriculture only	−0.758	−0.759	0.0005
Manufacturing only	−1.568	−1.564	−0.0045

Notes: Income changes are 100 times the log ratio of real income after the productivity change to baseline real income within each regime, in log percentage points. Without the system denotes noncooperation. Change in value is the difference between the two income changes, calculated before rounding. Aggregation uses fixed baseline-income weights across the 34 named economies. The main experiment changes agriculture and manufacturing productivity together. Component experiments change productivity only in the indicated sector. Tariffs are solved in each state. Component effects need not sum to the combined effect.

expenditure weights to define this common component gives a decline of 2.36 log percentage points. This normalization separates common from relative productivity changes, but does not explain why the latter have little effect on institutional value. The sectoral experiments provide evidence on that question.

Agricultural productivity changes range from approximately −24 to 20 percent, but the sector accounts for only 3.5 percent of modeled trade. These changes alone barely affect aggregate institutional value. Manufacturing changes are smaller, ranging from approximately −3.8 to 0.3 percent, yet affect a much larger trade base and produce most of the aggregate decline in value. A separate diagnostic confirms the importance of existing manufacturing tariff commitments. Without cooperation, manufacturing ceilings alone deliver 0.72 log percentage points of value. This compares with 0.73 when all tariff commitments are retained and 0.08 with agricultural ceilings alone (Appendix F.2). These stand-alone values are not additive. High observed agricultural tariffs therefore need not imply a large agricultural contribution to institutional value. The solved system also already selects low agricultural tariffs, leaving limited scope for further reductions.

Exposure and tariff adjustment partly offset each other. At each regime's initial tariffs, the productivity changes lower institutional value by 0.0080 points, while subsequent tariff responses restore 0.0046 points (Appendix Table F2). Both contributions are small relative to baseline value and the climate income loss.

6.2 National outcomes

National differences reflect how climate change alters the benefits of access to foreign goods and markets. Low tariffs can cushion domestic productivity losses by providing access to less affected foreign suppliers. Trade also transmits foreign productivity changes through import prices and export demand. How these forces change the value of restraint depends on the location of productivity changes, countries' trade patterns, and tariff responses.

Institutional value falls for 20 of the 32 optimizing governments and rises for 12 (Appendix Table F2). Australia has the largest proportional decline, 8.5 percent of its baseline value, although its own manufacturing productivity is virtually unchanged. The global manufacturing-

only experiment produces a decline close to the full climate result, indicating the importance of foreign manufacturing changes. Turkey and Ukraine, whose manufacturing productivity falls by about 4 percent, instead experience increases in institutional value of approximately 5 percent of their baseline values. These patterns are consistent with the cushioning and transmission effects of openness, although the experiments do not separately identify import-price and export-demand channels. Most of the change in these three economies arises before tariffs adjust, showing that productivity changes alter what existing restraint protects even without substantial policy responses.

Cooperation supplies no separate motive to compensate economies for climate damage. A partner's loss does not increase the fixed weight on its real income. The resulting tariff responses therefore need not favor the economies that lose most, and greater climate vulnerability need not imply a larger increase in institutional value.

6.3 Reforms of tariff restraint

A supplementary exercise asks whether future forms of the two restraints could mitigate climate-related income losses. Six stylized reforms tighten agricultural tariff commitments, strengthen cooperation, or combine the two, assuming agreement among participating governments (Appendices F.3 and F.4). Additional mitigation is the reform's climate-state income gain less its gain without climate change. All six provide negligible additional aggregate mitigation, at most 0.08 percent of the projected loss. The agricultural standstill caps already covered agricultural tariffs at their solved baseline system rates. The other five reforms lower the weighted-average climate-state income of the ten optimizing economies with the largest proportional climate losses under the existing system, although some individual members gain. All six provide negative additional mitigation for the agricultural-exposure group, defined as economies whose agricultural productivity falls by more than 15 percent.

Trade can support adaptation through changes in suppliers and production, which occur in every regime here. Restraints can provide additional protection when a shock induces tariff increases, as in the conflict exercise. Under climate change, reported sectoral mean tariffs move by less than a quarter of a percentage point (Appendix Table D2), and low agricultural tariffs under the system leave limited scope for further cuts. Stronger cooperation mainly delivers gains also present without climate change. Tariff removal changes prices and sacrifices revenue from affected imports, so improved access need not raise income, as Pakistan illustrates. Nath (2025) finds adaptation gains from broader trade-cost reductions that facilitate food imports and labor reallocation under subsistence demand. Fixed expenditure shares exclude the mechanism arising from subsistence food needs, and the reforms leave nontariff trade costs unchanged.[5] The small additional gains concern these reforms, rather than trade-based adaptation generally.

[5]The model omits export restrictions, domestic support, and transfers. For export restrictions and food-price volatility, see Martin and Anderson (2012).

7 Sensitivity and Robustness

This section examines how the welfare comparisons change with the welfare weight, tariff ceilings, trade elasticities, and the inputs to the conflict and climate applications. It distinguishes changes in estimated magnitudes from changes in the main conclusions.

The welfare weight matters mainly for how much of the system's value cooperation alone preserves. Across the common-weight specifications in Appendix Table G1, from the illustrative value of 0.26 to the upper importer-clustered confidence endpoint of 0.63, the value of the system ranges from 0.77 to 0.82 log percentage points. Over the same range, cooperation alone provides 69 to 102 percent of that value at the corresponding weight (Appendix Table G1). At the upper endpoint, cooperation alone is worth more than the system. The large overlap at the baseline is therefore conditional on the estimated weight and does not imply a general ranking of the restraints.

Ceiling specification also affects inference. Removing preferential ceilings lowers baseline value at the original welfare weight but raises the re-estimated weight from 0.50 to 0.88 (Appendix Table G2). This is consistent with attributing more observed restraint to government objectives when fewer constraints are modeled. However, the estimating sample also changes, so the change in the estimated weight does not isolate the restraint supplied by preferential tariff commitments.

The conflict-induced increase in the value of the remaining system to third countries remains within 11 percent of the principal estimate across the reported input, weight, and institutional checks (Appendix Table G3). World value continues to rise, while United States and Chinese values decline. Setting the two United States–China welfare weights to zero changes the third-country increase in the remaining system's value little but reduces its cooperation-only counterpart by about three quarters (Appendix Table G4). Adjustment accounts for 98 percent of the third-country increase under the main calculation order and 96 percent under the reverse order. The United States adjustment component instead changes sign. These are accounting allocations rather than uniquely identified causal contributions. Total value changes are unchanged (Appendix E.5).

The climate-induced change remains small across the reported specifications, although its sign varies across projection sources (Appendix Table G5). Its magnitude is at most approximately 1.5 percent of the corresponding baseline value. Changing trade elasticities by 20 percent, with the welfare weight re-estimated, moves baseline value between 0.68 and 0.99 log percentage points without changing this conclusion. These exercises assess the specified projections within the model, rather than uncertainty about future climate and economic development as a whole.

Accepted solutions pass competitive-equilibrium, projected-gradient, and unilateral-deviation search checks (Appendix B.4). Aggregate welfare comparisons vary little across accepted starting profiles, but these checks establish neither uniqueness nor global optimality.

8 Conclusion

This paper measures the welfare value of the world trading system as the real income protected by cooperation in government objectives and tariff commitments in feasible tariff choices. Separating these restraints distinguishes government preferences from policy constraints and permits their value to be evaluated as trading conditions change. Estimation from positive interior tariffs yields a foreign welfare weight of about one half. In the calibrated 2017 tradable economy, the two restraints together raise world real income by about 0.81 percent relative to noncooperation. At the estimated weight, each alone preserves most of this gain. The United States–China conflict lowers world income but raises the value of the remaining system for third countries by about 12 percent. Projected climate change leaves aggregate value nearly unchanged despite substantial income losses, while national values move in both directions. The six reforms studied provide negligible additional aggregate mitigation of climate-related income losses.

Three policy implications follow, conditional on the model and counterfactuals studied. First, a bilateral dispute can strengthen the welfare case for preserving agreements on other trade relationships. In the conflict experiment, unrestrained tariff responses reverse third countries' initial aggregate gain from diverted trade. The increase in value for every modeled third country supports maintaining mutual restraint, but does not establish each government's unilateral incentive to retain its own bindings. Second, tariff commitments alone generate approximately the same increase in third-country value as the system, while cooperation alone generates less than half. This comparison supports preserving the tariff ceilings that limit additional protection in this episode. Third, a reform's contribution to climate adaptation should be measured net of the gains it would deliver without climate change. The negligible additional mitigation found here motivates examining transfers, disciplines on export restrictions, and the broader trade-cost reductions studied by Nath (2025). It does not rule out benefits from targeted tariff reforms.

The analysis takes welfare weights and tariff ceilings as given and does not explain their formation, enforcement, or renegotiation. Its three-sector model omits intermediate inputs, nontariff policies, uncertainty, and political-economy motives beyond the estimated weights. One extension would model the choice of ceilings, drawing on Amador and Bagwell (2013), to connect their welfare value to institutional design. Another would introduce intermediate inputs as in Caliendo and Parro (2015), allowing production linkages to alter sectoral tariff incentives and the income protected by restraint. These extensions would connect the valuation of specified agreements to their design in a changing trading environment.

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A Competitive equilibrium and hat algebra

This appendix sets out the multi-sector Ricardian equilibrium, its representation in relative changes, and the derivatives used to evaluate tariff choices. The model follows Eaton and Kortum (2002) and the sectoral structure of Caliendo and Parro (2015), with tariff revenue rebated and no intermediate inputs. The relative-change formulation follows Dekle et al. (2008).

Let $\mathbf{T}$ denote productivity, C the set of imposed tariff coordinates, $\mathbf{q}$ their rates, and $\mathbf{x}$ the remaining tariffs. The full tariff vector is $\mathbf{t}(\mathbf{q}, \mathbf{x})$, with fixed background policies suppressed, and the environment is $e = (\mathbf{T}, \mathbf{q}; C)$. The main-text environments $r = 0, 1$ correspond to e_0, e_1 with the same C. In the climate application C is empty.

A.1 The competitive equilibrium

Countries, sectors, tariffs, and welfare use the notation of Section 2.1. Country i has labor $L_i > 0$, which moves freely across domestic sectors and is immobile internationally. Production has constant returns and uses labor alone, so every active sector pays the same wage w_i.

Households allocate the positive expenditure share α_{is} to sector s, with shares summing to one. Within each sector, CES preferences have elasticity $\rho_s > 1$ over a unit continuum of varieties. If P_{is} is the sectoral price index, sectoral expenditure is $\alpha_{is} I_i$ and the aggregate price index is $P_i = \prod_s (P_{is}/\alpha_{is})^{\alpha_{is}}$.

Each variety's labor productivity in country j and sector s is an independent Fréchet draw with scale $T_{js} > 0$ and shape $\theta_s > \rho_s - 1$. Multiplying T_{js} by c^{θ_s} for any $c > 0$ scales every draw by c. Delivery from j to i requires $d^s_{ji} \geq 1$ units per unit delivered, with $d^s_{ii} = 1$ and $t^s_{ii} = 0$. Including the tariff t^s_{ji}, the trade-cost factor and destination competitiveness index are

$$\tau^s_{ji} = d^s_{ji}(1 + t^s_{ji}), \qquad \Phi_{is} = \sum_j T_{js}(w_j \tau^s_{ji})^{-\theta_s}. \tag{A.1}$$

Sourcing each variety from the cheapest supplier gives the expenditure shares and sectoral price indices

$$\begin{aligned} \pi^s_{ji} &= \frac{T_{js}(w_j \tau^s_{ji})^{-\theta_s}}{\Phi_{is}}, \\ P_{is} &= A_s \Phi_{is}^{-1/\theta_s}, \qquad A_s = \left[\Gamma\left(1 + \frac{1-\rho_s}{\theta_s}\right)\right]^{1/(1-\rho_s)}, \end{aligned} \tag{A.2}$$

where π^s_{ji} is country j's share of country i's sectoral expenditure. Let M^s_{ji} denote tariff-exclusive sales from j to i. Producer receipts and tariff revenue are

$$M^s_{ji} = \frac{\alpha_{is} I_i \pi^s_{ji}}{1 + t^s_{ji}}, \qquad R_i = \sum_{j,s} t^s_{ji} M^s_{ji}. \tag{A.3}$$

Producer receipts include payment for labor used to produce goods lost in transit. With tariff revenue rebated and no international transfers, household budgets and labor-market clearing

require

$$I_i = w_i L_i + R_i,$$
$$w_j L_j = \sum_{i,s} M_{ji}^s. \tag{A.4}$$

Sectoral employment is $L_{js} = \sum_i M_{ji}^s / w_j$. Together, the budget and market-clearing conditions imply balanced trade at tariff-exclusive values. A competitive equilibrium for $(\mathbf{T}, \mathbf{t})$ consists of positive wages and expenditures satisfying (A.2) and (A.3)–(A.4), with one wage normalized to one. Policy regimes affect this equilibrium through their tariff schedules.

A.2 Equilibrium in relative changes

Let superscript 0 denote the baseline and write $\widehat{x} = x/x^0$ for each strictly positive variable. Expenditure shares, labor endowments, and iceberg costs remain fixed. The baseline tariff schedule need not be a policy equilibrium. The changes in trade costs and competitiveness are

$$\widehat{\tau}_{ji}^s = \frac{1 + t_{ji}^s}{1 + t_{ji}^{s,0}}, \qquad \widehat{\Phi}_{is} = \frac{\Phi_{is}}{\Phi_{is}^0} = \sum_j \pi_{ji}^{s,0} \widehat{T}_{js} (\widehat{w}_j \widehat{\tau}_{ji}^s)^{-\theta_s}. \tag{A.5}$$

For simultaneous changes in technologies and tariffs, sourcing shares and price indices satisfy

$$\widehat{\pi}_{ji}^s = \frac{\widehat{T}_{js} (\widehat{w}_j \widehat{\tau}_{ji}^s)^{-\theta_s}}{\widehat{\Phi}_{is}},$$
$$\widehat{P}_{is} = \widehat{\Phi}_{is}^{-1/\theta_s}, \qquad \widehat{P}_i = \prod_s \widehat{P}_{is}^{\alpha_{is}}.$$

Tariff revenue as a share of expenditure is

$$r_i = \frac{R_i}{I_i} = \sum_s \alpha_{is} \sum_j \pi_{ji}^s \frac{t_{ji}^s}{1 + t_{ji}^s},$$

with $0 \le r_i < 1$ for finite nonnegative tariffs, so the household budget gives

$$I_i = \frac{w_i L_i}{1 - r_i}, \qquad \widehat{I}_i = \widehat{w}_i \frac{1 - r_i^0}{1 - r_i},$$

which is well defined at zero baseline revenue. With the baseline producer-sales weights

$$\omega_{ji}^{s,0} = \frac{M_{ji}^{s,0}}{w_j^0 L_j}, \qquad \sum_{i,s} \omega_{ji}^{s,0} = 1,$$

receipts and labor-market clearing become

$$\widehat{M}^s_{ji} = \widehat{\pi}^s_{ji}\widehat{I}_i \frac{1+t^{s,0}_{ji}}{1+t^s_{ji}},$$
$$\widehat{w}_j = \sum_{i,s} \omega^{s,0}_{ji}\widehat{M}^s_{ji}. \tag{A.6}$$

Equations (A.5)–(A.6), with one $\widehat{w}_j = 1$, determine the equilibrium in relative changes on the chosen branch, and welfare satisfies

$$\widehat{W}_i = \frac{\widehat{I}_i}{\widehat{P}_i} = \widehat{w}_i \frac{1-r^0_i}{1-r_i} \prod_s \widehat{\Phi}^{\alpha_{is}/\theta_s}_{is}.$$

The system requires baseline bilateral shares, producer-sales weights, expenditure shares, tariffs, and trade elasticities. Zero baseline trade corresponds to prohibitive trade costs and remains zero, so trade-flow hats apply only to positive baseline cells. At fixed baseline prices, the government objective becomes

$$P^0_i W_i + \sum_{j\neq i} \lambda_{ij} P^0_j W_j = I^0_i \widehat{W}_i + \sum_{j\neq i} \lambda_{ij} I^0_j \widehat{W}_j,$$

so baseline expenditures and the cooperation weights suffice to aggregate welfare changes in the policy objective.

A.3 Equilibrium derivatives

Government tariff choices use welfare derivatives that account for market clearing. Normalize $w_N = 1$ and collect the remaining log wages and all log expenditures in

$$\boldsymbol{\xi} = (\log w_1, \ldots, \log w_{N-1}, \log I_1, \ldots, \log I_N).$$

Let $\boldsymbol{\mu}$ collect log technologies and tariff levels. Substituting sourcing shares into the expenditure and labor-market equations gives the square system

$$\mathbf{F}(\boldsymbol{\xi}, \boldsymbol{\mu}) = \mathbf{0}.$$

It contains the N expenditure equations $(1-r_i)I_i - w_i L_i = 0$ and the $N-1$ labor-market equations $w_j L_j - \sum_{i,s} M^s_{ji} = 0$, $j < N$. The omitted labor-market equation follows from their sum and the tariff-revenue identity.

If $\mathbf{F}$ is continuously differentiable and $\mathbf{F}_{\boldsymbol{\xi}}$ is nonsingular, the implicit-function theorem gives a locally unique, differentiable equilibrium branch. Along that branch,

$$\frac{\partial \boldsymbol{\xi}}{\partial \mu_k} = -\mathbf{F}^{-1}_{\boldsymbol{\xi}} \mathbf{F}_{\mu_k}, \qquad \frac{d \log W_i}{d\mu_k} = \frac{\partial \log W_i}{\partial \mu_k} - \frac{\partial \log W_i}{\partial \boldsymbol{\xi}} \mathbf{F}^{-1}_{\boldsymbol{\xi}} \mathbf{F}_{\mu_k}. \tag{A.7}$$

Partial derivatives on the right hold $\boldsymbol{\xi}$ fixed. The total derivative adds the induced changes in wages, expenditures, and tariff revenue. These equilibrium responses determine tariff incentives.

The nonsingularity condition ensures local regularity of the selected competitive equilibrium. A differentiable policy equilibrium additionally requires regularity of the policy system. Neither condition establishes global uniqueness.

A.4 Common productivity scaling

Corollary 1 (Common productivity scale)**.** *Fix the imposed-coordinate set, imposed tariffs, remaining policy domains, welfare weights, baseline prices, and all nontechnology primitives. For $c > 0$, let $T_{js}(c) = c^{\theta_s} T_{js}$ for every country and sector. Corresponding equilibrium selections have unchanged tariff profiles and $W_i^R(\mathbf{T}(c), \mathbf{q}; C) = cW_i^R(\mathbf{T}, \mathbf{q}; C)$. Every institutional log value is unchanged.*

Proof. At any fixed tariff profile, scaling technology as specified multiplies Φ_{is} by c^{θ_s}. Sourcing shares are unchanged at the original normalized wages. Expenditures, tariff revenues, and labor-market clearing therefore remain unchanged, while every sectoral price index is divided by c. Cobb–Douglas shares sum to one, so P_i is divided by c and W_i is multiplied by c. Each government objective is consequently multiplied by the same positive scalar at every feasible profile. Neither imposed tariffs nor remaining domains change. Best-response correspondences and the policy-equilibrium set are unchanged. Pairing each equilibrium with its scaled counterpart yields equal log welfare gaps across regimes. □

This homogeneity property separates a common productivity change from changes in relative productivity. Choose fixed reporting weights ϖ_{js} that sum to one and write

$$h = \sum_{j,s} \varpi_{js} \log \widehat{a}_{js}, \qquad \widetilde{h}_{js} = \log \widehat{a}_{js} - h.$$

Scaling the relative-change vector by e^h recovers the full productivity change. Corresponding equilibrium selections have identical policies and institutional log values, while income levels differ by e^h. This holds for any fixed reporting weights and leaves the effects of relative productivity changes to quantitative analysis.

B Policy equilibria and welfare accounting

This appendix states policy optimality conditions, defines welfare aggregation and decomposition, and documents numerical verification.

B.1 Existence and optimality conditions

Fix the environment e and a selection of competitive equilibria. A pure-strategy conditional policy equilibrium exists if each remaining policy domain is nonempty, compact, and convex, welfare is continuous, and each government's objective is quasi-concave in its own choices.

The quantitative analysis does not establish these sufficient conditions. It evaluates candidate equilibria using stationarity conditions and searches for profitable unilateral deviations.

For a remaining coordinate $x_{i\ell} \in [l_{i\ell}, h_{i\ell}]$ with positive domain width, necessary conditions are

$$\begin{cases} G_{i,x_{i\ell}} = 0, & l_{i\ell} < x_{i\ell} < h_{i\ell}, \\ G_{i,x_{i\ell}} \leq 0, & x_{i\ell} = l_{i\ell}, \\ G_{i,x_{i\ell}} \geq 0, & x_{i\ell} = h_{i\ell}. \end{cases} \tag{B.1}$$

The derivative includes the competitive-equilibrium response. Concavity over the government's entire policy domain would make these conditions sufficient for a global best response. A singleton domain has no feasible direction and hence supplies no directional first-order restriction. Imposed coordinates likewise supply no estimating restriction.

B.2 Welfare aggregation and level reporting

Write $W_i^R(e)$ and $v_i^R(e)$ for equilibrium income and regime value in environment e, as defined in Section 2. For a given remaining tariff profile, let $u_i(e, \mathbf{x}) = \log W_i(\mathbf{T}, \mathbf{t}(\mathbf{q}, \mathbf{x}))$.

For a reporting group $\mathcal{G}$, aggregate institutional value is

$$\mathcal{V}_{\mathcal{G}}^R(e) = 100 \sum_{i \in \mathcal{G}} \omega_i^0 v_i^R(e), \qquad \omega_i^0 = \frac{I_i^0}{\sum_{j \in \mathcal{G}} I_j^0}. \tag{B.2}$$

The weights use income in the calibrated allocation and remain fixed across environments and regimes. World results cover the 34 named economies. Third-country results cover the 32 named economies other than the United States and China, with the income weights rescaled to sum to one over that group. The index averages national log real-income gains and is reported in log percentage points. A complementary level measure is

$$L_{\mathcal{G}}^R(e) = \sum_{i \in \mathcal{G}} P_i^0 \left[W_i^R(e) - W_i^{\mathrm{nc}}(e) \right].$$

This level measure scales with common productivity, unlike institutional log value. It reports income in the calibrated tradable economy at baseline prices. Production data measure gross output, so dollar amounts are not changes in world GDP.

B.3 Finite and local decompositions

The finite decomposition in (4) first changes the environment and then adjusts remaining tariffs, assigning their interaction to adjustment. This accounting sequence does not impose a policy timeline or require a differentiable equilibrium path. Subtracting the noncooperative components gives the change in institutional value in (5).

Policy adjustment can also be divided into partner and own components. With partners

adjusted first,

$$A_i^{\text{par},R} = u_i(e_1, \mathbf{x}_i^{R,0}, \mathbf{x}_{-i}^{R,1}) - u_i(e_1, \mathbf{x}_i^{R,0}, \mathbf{x}_{-i}^{R,0}),$$
$$A_i^{\text{own},R} = u_i(e_1, \mathbf{x}_i^{R,1}, \mathbf{x}_{-i}^{R,1}) - u_i(e_1, \mathbf{x}_i^{R,0}, \mathbf{x}_{-i}^{R,1}).$$

Each mixed profile requires a new competitive equilibrium. In the conflict application, both steps retain the imposed tariffs at $\mathbf{q}_1$, and the first holds only the country's remaining choices fixed. The components sum to A_i^R, and their differences across regimes decompose the adjustment contribution to value. Reversing the own/partner order can change these components without changing total adjustment. Reversing the exposure/adjustment order can instead change both broad components without changing total value.

For the local counterpart, let $e(z)$ be a path with a fixed imposed-coordinate set C and write

$$u_i(z, \mathbf{x}) = u_i(e(z), \mathbf{x}) = \log W_i(\mathbf{T}(z), \mathbf{t}(\mathbf{q}(z), \mathbf{x})).$$

Along differentiable equilibrium branches,

$$\frac{dv_i^R}{dz} = \underbrace{u_{i,z}(z, \mathbf{x}^R) - u_{i,z}(z, \mathbf{x}^{\text{nc}})}_{\text{difference in exposure}} + \underbrace{\sum_j \left[\nabla_{\mathbf{x}_j} u_i(z, \mathbf{x}^R) \cdot \frac{d\mathbf{x}_j^R}{dz} - \nabla_{\mathbf{x}_j} u_i(z, \mathbf{x}^{\text{nc}}) \cdot \frac{d\mathbf{x}_j^{\text{nc}}}{dz} \right]}_{\text{difference in policy adjustment}}. \tag{B.3}$$

The partial derivative $u_{i,z}$ holds the remaining tariff choices fixed but permits the imposed tariffs and productivity to change. It also includes the competitive-equilibrium response. For climate change it is exposure to a productivity shift. For the conflict it is exposure to the imposed bilateral tariff changes.

The identity follows from the chain rule. With a locally unchanged set of active constraints, a nonsingular Jacobian for the remaining policy system gives a differentiable branch under the usual smoothness conditions. At an interior noncooperative choice, $\partial u_i / \partial x_{i\ell} = 0$. Under cooperation, the same first-order condition gives

$$\frac{\partial u_i}{\partial x_{i\ell}} = -\sum_{j \neq i} \lambda_{ij} \frac{P_j^0 W_j}{P_i^0 W_i} \frac{\partial u_j}{\partial x_{i\ell}}.$$

The domestic gradient need not vanish because the government also values foreign income. A choice at a fixed bound has zero local adjustment while it remains there. These arguments concern $\mathbf{x}_i$ only. A moving imposed tariff in $\mathbf{q}$ remains part of the external exposure term.

B.4 Numerical solution and verification

Projected Jacobi–Newton iteration solves the stacked first-order and bound conditions. Government gradients are evaluated at a common tariff profile, and numerical curvature approximations with adaptive damping determine simultaneous updates. Competitive equilibrium and welfare derivatives are recomputed after each update. Finite differences from independently re-solved equilibria check the derivatives.

Acceptance requires a competitive-equilibrium residual below 10^{-12} and a largest projected relative objective gradient below 10^{-6}. The gradient uses log-gross-tariff coordinates and excludes infeasible directions. Bound contact uses a tolerance of 10^{-7} in tariff fractions. Projected quasi-Newton searches hold partner tariffs fixed and reject a candidate if a searched relative objective gain exceeds 10^{-6}. All reported solutions satisfy these checks, which establish neither global optimality nor uniqueness.

The unrestricted baseline starts from observed tariffs and the climate state from its corresponding baseline solution. Both are also solved from lower-bound schedules in all four regimes. Across these accepted candidates, tariff vectors differ by up to 23.63 tariff percentage points and country welfare by 0.00329 log percentage points. For the principal conflict games, the corresponding maxima are 18.2866 tariff points and 0.005265 log income points. For the five broader climate reforms, world income at each productivity state differs by less than 0.00005 log percentage points across accepted starts. Table B1 reports the principal applications' value-change ranges when each endpoint is selected independently. Welfare comparisons are stable across these candidates, although tariff differences may reflect numerical tolerance or equilibrium multiplicity.

Table B1: Accepted-candidate ranges for changes in system value

Population	Minimum	Maximum
Panel A. Conflict		
World	0.063631	0.063686
Third countries	0.144911	0.145028
United States	−0.082190	−0.082042
China	−0.003404	−0.003324
Panel B. Climate		
World	−0.003433	−0.003390

Notes: Entries are changes in the full system's real-income gain over noncooperation between each application's initial and changed environments, in log percentage points. Each range permits independent selection of accepted system and noncooperative candidates at both endpoints. The climate panel covers sixteen combinations. These are numerical candidate ranges, not statistical confidence intervals.

The unrestricted noncooperative game optimizes all 2,989 bilateral-sector tariffs separately, including observed preferential zeros. Among the 1,021 importer–exporter pairs with at least two eligible sectors, the largest baseline within-pair tariff range is 0.0101 tariff percentage points. The candidate starts from nonuniform observed tariffs. The accepted solution from zero active tariffs differs by at most 0.0016 tariff percentage points. This supports the numerical consistency of the tariff structure discussed in Appendix D.2.

Imposed and background tariff coordinates remain fixed, and income changes and finite decompositions reconcile to numerical precision. The common-productivity check retains baseline tariffs after a common scale change and applies the same optimality and deviation checks. Removing the common component of the climate shock reproduces policies and institutional log values on corresponding equilibrium selections, as Corollary 1 requires.

C Data, calibration, and estimation

This appendix documents the data construction, accounting reconciliation, tariff domains, and estimation checks.

C.1 Sources and parameters

Trade data are BACI's 2017 HS six-digit records in the HS2017 nomenclature. Products are assigned to sectors through CPC 2.1 and ISIC Revision 4. Tariff concordances retain the first target in one-to-many mappings and use a four-digit-prefix match for otherwise unmatched codes. HS2002 schedules pass through the supplied HS2007–HS2002 correlation and then the HS2007-to-HS2017 mapping. Data-coverage shares use raw trade, while estimation shares use the balanced baseline described in Appendix C.2.

Applied tariffs come from WITS MFN and preferential schedules for the nearest available year to 2017. All EU-28 members use the common 2017 EUN schedule. An available ad-valorem-equivalent MFN schedule replaces the plain MFN schedule for the same year and nomenclature. After expanding beneficiary groups and applying exclusions, the construction selects the lowest finite eligible program average. A lower MFN average replaces it only for the same reporter, exporter, source year, nomenclature, and source product.

Identical duplicates receive no additional weight, and conflicting observations within a program stop construction. Distinct source products mapping into one HS2017 product receive equal weights, including comparable MFN-only components. These rates summarize available schedules rather than preference utilization. Non-2017 rates cover 4.22 percent of active imports, including 0.50 percentage points from the sole applied-tariff supplement. That supplement assigns a 2019 crude-oil ad-valorem equivalent of 2.4 percent to 36 missing United States observations, covering 36.9 percent of its extraction imports. The proxy is excluded from observed product-rate support when measuring interior coverage.

Agricultural output uses FAOSTAT crop and livestock production values. Extraction and manufacturing use UNIDO gross output, supplemented by OECD TiVA 2025 observations for 2017. Taiwan retains available UNIDO observations, and aggregate totals are not added to their constituents. Rest of world closes the model and is excluded from group summaries. Agriculture accounts for 3.5 percent of calibrated tariff-exclusive cross-border imports, including rest-of-world flows. Advanced economies are the World Bank's 2017 high-income economies, with Taiwan also treated as advanced.

Table C1: Industry elasticities and aggregation weights

Industry	ISIC divisions	Elasticity	Sector weight
Agriculture, forestry, and fishing	01–03	8.11	1
Mining and quarrying	05–09	15.72	1
Food, beverages, and tobacco	10–12	2.62	3/24
Textiles and apparel	13–14	5.60	2/24
Leather*	15	5.00	1/24
Wood	16	10.83	1/24
Paper and printing	17–18	9.07	2/24
Coke and refined petroleum*	19	15.00	1/24
Chemicals and pharmaceuticals	20–21	4.75	2/24
Rubber and plastics	22	1.67	1/24
Nonmetallic minerals	23	2.76	1/24
Basic and fabricated metals*	24–25	5.81	2/24
Computers, electronics, and electrical equipment	26–27	10.60	2/24
Machinery	28	1.52	1/24
Transport equipment	29–30	1.84	2/24
Furniture, other manufacturing, and repair/installation	31–33	5.00	3/24

Notes: Elasticities are dimensionless. Asterisks identify the calibration assignments stated in the text. Other entries follow Caliendo and Parro (2015). Each ISIC Revision 4 division receives its industry's elasticity. An industry's sector weight is its share of divisions in that model sector, and the weights sum to one within each sector.

Industry trade elasticities follow Caliendo and Parro (2015), with calibration assignments of 5.00 for leather, 5.81 for basic and fabricated metals, and 15.00 for coke and refined petroleum. These three assignments are distinguished from the source estimates in Table C1. Equal weights across ISIC Revision 4 two-digit divisions give sector elasticities of 8.11 for agriculture, 15.72 for extraction, and 5.62 for manufacturing.

C.2 Accounting reconciliation and the balanced baseline

Trade and production are mapped into tariff-exclusive sales $M_{ji}^{s,0}$, production B_{js}^0, and tariff-inclusive expenditure E_{is}^0. Domestic sales are production less exports, $M_{jj}^{s,0} = B_{js}^0 - \sum_{i \neq j} M_{ji}^{s,0}$. The accounts satisfy

$$E_{is}^0 = \sum_j (1 + t_{ji}^{s,0}) M_{ji}^{s,0}, \qquad Y_i^0 = \sum_s B_{is}^0, \qquad I_i^0 = \sum_s E_{is}^0,$$
$$R_i^0 = \sum_{j \neq i,s} t_{ji}^{s,0} M_{ji}^{s,0}, \qquad D_i^0 = I_i^0 - Y_i^0 - R_i^0, \qquad \alpha_{is} = E_{is}^0 / I_i^0.$$

Thus $\pi_{ji}^{s,0} = (1+t_{ji}^{s,0}) M_{ji}^{s,0} / E_{is}^0$ and $\omega_{ji}^{s,0} = M_{ji}^{s,0} / Y_j^0$. The model treats production as labor income, $Y_i^0 = w_i^0 L_i$, while the data measure gross output rather than value added or compensation.

When exports exceed 99.9 percent of available country-sector output, output is raised until domestic sales equal at least 0.1 percent of production. This adds $313.3 billion across seven cells, or 0.62 percent of completed output, mainly in rest-of-world and Iranian extraction. The adjustment reconciles the accounts but does not recover missing output. Agricultural sources omit forestry and fishing and lack Taiwan and the United Arab Emirates. Other adjusted cells include agriculture in Norway and Singapore and extraction in Singapore. Rest-of-world

production has missing constituent sources in all three sectors.

European Union internal shipments count as domestic trade, and re-exports remain as recorded. Hong Kong is included in rest of world. Raw deficits are removed at observed tariffs and unchanged technologies using the relative-change calculation of Dekle et al. (2008). During this calculation, the household budget in Appendix A.2 becomes $I_i = w_i L_i + R_i + D_i$. Wages, expenditure, sourcing shares, and price indices adjust jointly to a baseline with $D_i = 0$.

C.3 Ceilings and estimation-sample construction

WTO ceilings target the end of 2017. Verified national-line stages replace downloaded final duties before aggregation. Where verification is unavailable, a supplied historical-base duty is used, or otherwise the downloaded final duty serves as a proxy. Rates are not interpolated between implementation dates. The construction uses 4,571 verified-stage records, 1,399 historical-base proxies, and 255,339 final-bound proxies. These provenance classifications do not establish a complete reconstruction of legal ceilings in 2017.

National-line rates are averaged within mapped HS6 products and concorded to HS2017. A product with any explicitly unbound line is treated as unbound. Replication records retain the historical overrides and their supporting sources. Schedules cover 31 of the 33 WTO members in the sample. Saudi Arabia and the United Arab Emirates lack consolidated schedules and belong to a partly represented customs union. They and rest of world retain observed tariffs. Iran has no WTO ceiling and optimizes under the pooled welfare weight but is excluded from estimation.

Bound products enter at the smaller of their bound rate and the numerical cap. Explicitly unbound products enter at the cap. Unknown bindings receive the cell's import-weighted average of known capped rates, or the cap if none is known. Unknown products account for 10.3 percent of active imports and 54.6 percent of Switzerland's imports. Approximate matches can still be classified as known. After aggregation, ceilings are raised to observed tariffs in 132 active cells covering 15.1 percent of active imports.

Preferential pairs between separately represented economies receive their observed cell rate as a ceiling. The operative classification is the `preferential` indicator in the replication file `Step_1_output/cell_diagnostics.csv`. It includes reciprocal agreements and unilateral schemes. These tariffs can fall but cannot enter the interior estimating sample. For rest-of-world exports, preferential source components enter the ceiling at their selected applied rates and other components follow the binding rules above. Preferential components cover 26.0 percent of imports from rest of world and are excluded from interior coverage at the same fixed weights.

The numerical upper bound is the larger of 200 percent and the observed tariff. Five Korean agricultural indices exceed 200 percent, ranging from 238.67 to 423.67 percent and covering 0.0093 percent of active imports. Their tariffs remain free to fall. Appendix C.4 gives the aggregation formulas, and Table G5 reports alternative domain specifications.

An interior cell has a positive tariff at least δ percentage points below its ceiling, following the tariff-water distinction in Beshkar et al. (2015) and Nicita et al. (2018). The baseline sets

Table C2: Cell classification and product-level coverage by economy

Economy	Interior	Adjustable zero	Fixed zero	Ceiling class	Mean interior coverage	Unknown binding share
Advanced economies						
AUS	7	2	40	51	0.435	0.049
CAN	7	2	21	72	0.213	0.051
CHE	4	0	26	70	0.000	0.546
CHL	31	0	35	28	0.978	0.000
EU	0	0	12	90	—	0.125
ISR	51	7	9	19	0.345	0.007
JPN	11	1	8	79	0.075	0.133
KOR	16	0	14	72	0.739	0.028
NOR	0	0	32	63	—	0.121
SGP	6	32	42	0	0.000	0.039
TWN	4	1	4	91	0.093	0.006
USA	19	0	16	67	0.137	0.080
Developing economies						
ARG	32	2	10	28	0.607	0.132
BGD	84	2	0	12	0.746	0.000
BRA	43	1	10	43	0.605	0.077
CHN	8	0	10	84	0.318	0.159
COL	58	0	2	31	0.923	0.001
IDN	65	3	10	23	0.532	0.114
IND	43	0	3	56	0.729	0.029
IRN	89	0	0	0	—	—
KAZ	0	1	7	80	—	0.150
MEX	33	11	13	21	0.479	0.046
MYS	86	7	0	6	0.222	0.161
PAK	73	0	0	20	0.996	0.000
PER	37	8	28	16	0.262	0.000
PHL	52	0	6	18	0.762	0.039
RUS	0	0	8	93	—	0.155
THA	86	7	2	7	0.194	0.105
TUR	44	0	12	42	0.395	0.084
UKR	2	2	6	63	0.500	0.006
VNM	3	0	5	70	0.218	0.000
ZAF	54	23	7	16	0.297	0.000
Total	1,048	112	398	1,431		

Notes: The first four numerical columns count positive-import government–partner–sector cells. Adjustable zeros can rise, whereas fixed zeros have singleton domains. The ceiling class contains positive tariffs less than five percentage points below the ceiling. Mean interior coverage is the unweighted mean, across an economy's interior cells, of each cell's product-level interior import share. Unknown binding share is the raw-import-weighted unknown-binding share across its active cells. Both shares are fractions between zero and one. Dashes denote no interior cells or, for Iran, inapplicable WTO-based coverage. Iran's 89 interiors are excluded from the 959-cell estimating sample.

$\delta = 5$, with two and ten used in sensitivities. This classification of observed cells for estimation remains fixed across counterfactual states. Equilibrium ceiling contact may change. Product-level interior coverage measures the cell's import share with observed positive rates below known bindings, excluding preferential rest-of-world components. Table C2 reports the classification and product-level coverage by economy.

The 2,989 instruments comprise 959 estimation interiors, 1,431 ceiling-class cells, 112 adjustable zeros, 398 fixed zeros, and Iran's 89 interiors. Of the fixed zeros, 383 are preferential and 15 nonpreferential. The broad equality regression uses the first three groups. Fixed zeros supply no feasible direction and enter neither equality regressions nor directional optimality

diagnostics.

C.4 Tariff aggregation and product-level coverage

Let $\phi^0_{jip} = M^0_{jip} / \sum_{p' \in s} M^0_{jip'}$ be fixed raw tariff-exclusive import weights. Write $\widetilde{t}^0_{jip}$ for the matched applied rate, with unmatched products assigned their cell's finite-rate weighted average. Let $\widetilde{b}_{jip}$ incorporate the binding, cap, missing-data, and rest-of-world rules above. For nonpreferential cells,

$$t^{s,0}_{ji} = \sum_{p \in s} \phi^0_{jip} \widetilde{t}^0_{jip}, \qquad \bar{t}^s_{ji} = \max \left\{ t^{s,0}_{ji}, \min \left[K, \sum_{p \in s} \phi^0_{jip} \widetilde{b}_{jip} \right] \right\}, \tag{C.1}$$

where K is the numerical cap. Preferential cells instead have $\bar{t}^s_{ji} = t^{s,0}_{ji}$. Unmatched applied rates cover 0.26 percent of active imports. The index preserves revenue at the completed product rates and fixed weights, but not price-index derivatives or revenue responses to changes in product composition.

An interior sectoral cell can contain products at zero tariffs or at their bindings. A feasible product-level adjustment could raise tariffs only where room remains and lower them only where rates are positive. The sectoral game does not allocate its changes across products. Lower estimates at higher interior coverage (Table 2) show that composition matters, but do not identify the direction or magnitude of aggregation bias.

C.5 Estimation diagnostics

The importer-clustered standard error is 0.066, giving a 95 percent interval of 0.371 to 0.628. Allowing dependence between cells sharing a country in either role raises the standard error to 0.158 and widens the interval to 0.191 to 0.809.

The instrumental-variable estimate is 0.408, with standard error 0.082 and clustered first-stage F statistic 153.9. Inverting the importer-clustered Anderson–Rubin statistic gives the 95 percent set $[0.227, 0.555]$ (Anderson and Rubin, 1949). Asymptotic critical values mean that coverage is not exact with a finite number of clusters.

Leave-one-importer estimates range from 0.469 without India to 0.532 without South Africa. Leave-one-exporter estimates range from 0.389 without rest of world to 0.696 without China. Importer fixed effects give 0.480, with standard error 0.069.

The estimated weight on advanced partners exceeds that on developing partners by 0.103, with clustered standard error 0.050. Among the sixteen governments with at least twenty interior cells, government-specific estimates range from -0.42 for Mexico to 1.05 for Peru, with twelve in $[0, 1]$. Their standard errors are clustered by exporter. The policy sensitivity clips these estimates to $[0, 1]$ and assigns the pooled weight to governments with fewer observations.

Without preferential ceilings, the least-squares estimate is 0.880, with standard error 0.089 and 95 percent interval $[0.705, 1.055]$. The IV estimate is 0.805. These differences reflect both

the changed treatment of constraints and a changed estimating sample. Table G2 separates the welfare effects of changing ceilings at a fixed weight from re-estimation.

Restricting interior coverage to at least one quarter and one half gives IV estimates of 0.187 and 0.065, with standard errors 0.098 and 0.108. Coverage weighting gives 0.190, with standard error 0.095. Among nine importers with at least twenty cells at all three thresholds, the mean importer estimate falls from 0.380 to −0.015. The cells change even within this common importer sample, so the comparison does not isolate aggregation error.

Treating zero tariffs through a censored likelihood gives 0.551, compared with 0.548 when they enter as equality conditions. The ceiling inequality holds on 83.6 percent of imports in the five-point ceiling class and 90.9 percent at observed ceiling contacts, defined by an applied-ceiling gap no greater than 10^{-10}. Singleton domains enter neither directional diagnostic.

D Additional baseline results

This appendix reports national baseline values, the tariffs underlying them, and the gains and spillovers from unilateral departure.

D.1 Aggregate and national values

Figure D1 compares observed tariffs and solved system tariffs against the same unrestricted noncooperative benchmark. Table D1 separates the national gains from cooperation and commitments. Their stand-alone values can exceed the full-system value for individual economies and need not add to it.

The level measure in Appendix B.2 gives a baseline full-system gain of approximately $391.9 billion in annual-equivalent modeled income at baseline prices. This amount concerns the calibrated tradable economy rather than world GDP.

D.2 Tariff profiles and binding commitments

Table D2 compares observed and solved tariffs. The system selects substantially lower agricultural tariffs than those observed, particularly in advanced economies. The noncooperative means vary little across sectors, although tariffs differ across exporters.

The ceiling-contact comparison covers 2,591 variable cells, including Iran's 89 cells, and excludes 398 zero-tariff singleton domains. Commitments alone produce 1,223 contacts, covering 82.4 percent of baseline imports in the eligible cells. The full system produces 857 contacts, covering 69.3 percent. Cooperation moves 366 commitments-only contacts below their ceilings and creates no new contact. All contacts are at institutional ceilings below numerical upper bounds. Contact alone does not establish a positive shadow value.

The unrestricted noncooperative tariffs are 18.195 percent on United States imports from China and 18.250 percent on Chinese imports from the United States in each sector. Unrestricted sector-uniform tariffs accord with Lashkaripour (2021, Appendix D). The result concerns

Table D1: Baseline value of each restraint by economy

Economy	Commitments only	Cooperation only	Full system
AUS	2.90	1.46	3.09
CAN	3.89	1.75	4.02
CHE	5.79	2.52	5.77
CHL	3.16	1.45	3.10
EU	0.08	0.68	0.21
ISR	2.79	1.88	2.85
JPN	0.41	0.77	0.54
KOR	1.70	1.65	1.74
NOR	5.44	1.55	5.56
SGP	3.49	2.41	3.77
TWN	1.50	1.45	1.63
USA	0.46	0.94	0.55
ARG	0.48	0.71	0.59
BGD	2.60	1.62	2.89
BRA	0.91	0.59	0.97
CHN	0.13	0.36	0.13
COL	2.03	0.93	2.01
IDN	0.75	0.65	0.84
IND	0.34	0.51	0.35
IRN	2.46	1.10	2.46
KAZ	3.12	1.38	3.20
MEX	6.36	2.21	6.37
MYS	4.34	2.10	4.23
PAK	0.89	0.58	1.03
PER	1.67	0.84	1.68
PHL	4.00	2.20	4.17
RUS	1.36	0.83	1.48
THA	2.38	1.63	2.46
TUR	1.39	1.07	1.47
UKR	1.73	1.21	2.08
VNM	1.64	1.42	1.86
ZAF	1.82	1.22	2.09

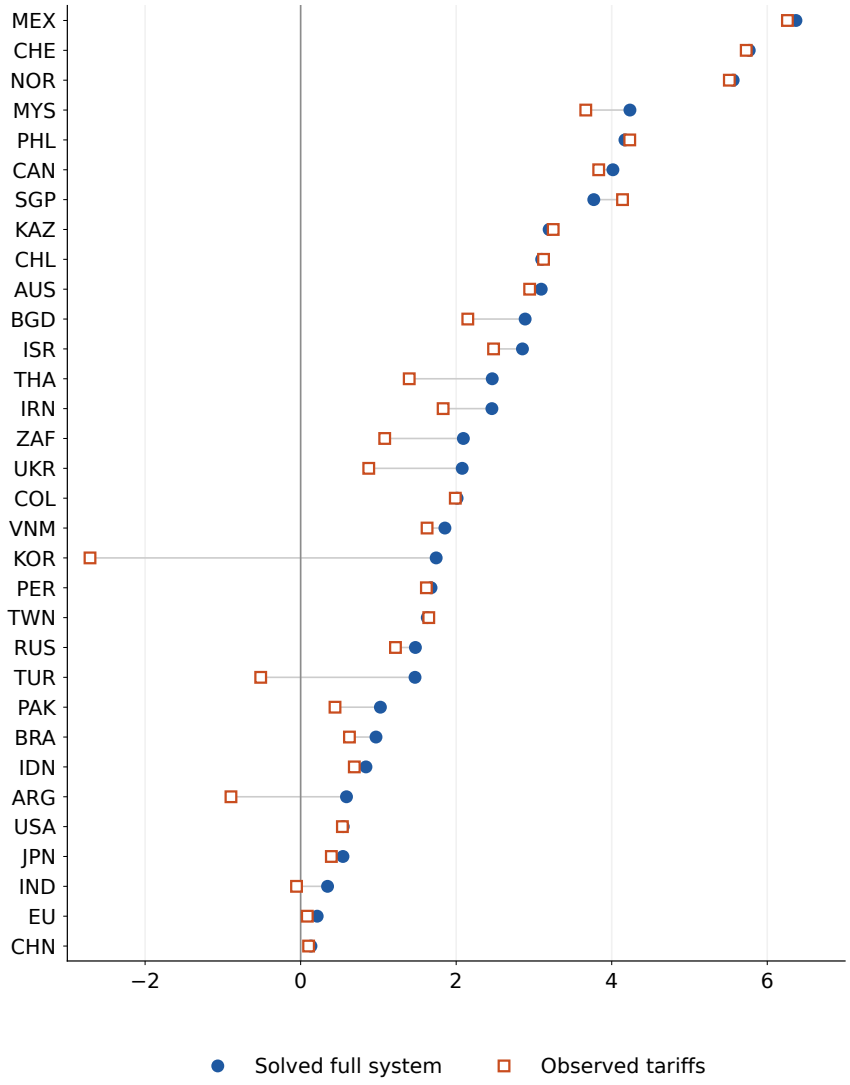
Notes: Entries are real-income gains under each restraint or the full system relative to unrestricted noncooperation at baseline productivity, in log percentage points. The table covers the 32 optimizing governments. Stand-alone values need not add to the full-system value, and a stand-alone value can exceed it.

Table D2: Mean tariffs by regime, group, and sector

Importer group	Sector	Observed	System	Commit.	Noncoop.
Advanced	agriculture	12.7	2.1	3.6	16.2
Advanced	extraction	0.6	0.6	1.1	16.3
Advanced	manufacturing	2.4	1.7	2.6	16.1
Developing	agriculture	7.9	4.2	8.0	15.4
Developing	extraction	2.4	3.5	6.6	15.5
Developing	manufacturing	6.3	4.0	7.7	15.3

Notes: Entries are unweighted mean ad valorem tariff rates, expressed in percent, at baseline productivity. Means cover cells with positive baseline trade in each importer group and sector. Every mean changes by less than a quarter of a percentage point under the climate shock.

Figure D1: National gains from the world trading system in 2017



Notes: The 32 optimizing governments are ordered by the value of the solved full system. Filled circles use the solved system tariffs, and open squares use observed tariffs. Both report $100 \log(W_i/W_i^{\mathrm{nc}})$, in log percentage points, relative to the same baseline noncooperative equilibrium. Negative values indicate lower modeled real income than under that benchmark. The rest of world and the two nonoptimizing named economies are excluded.

domestic-income maximization and permits different rates across exporters. A common wage links production costs within an exporter. Holding wages and total payments to each origin fixed, sectoral tariff differences distort purchases without providing an additional terms-of-trade gain. Binding ceilings and imposed tariffs alter this policy problem. Appendix B.4 verifies the numerical tariff structure.

D.3 Unilateral departure and partner spillovers

A unilateral departure removes one government's institutional ceilings and foreign welfare weights while fixing partner tariffs at their system values. Its domestic-income gain is

$$g_i = \max_{\mathbf{t}_i \in \mathcal{D}_i} \log W_i(\mathbf{T}, \mathbf{t}_i, \mathbf{t}_{-i}^{\text{sys}}) - \log W_i(\mathbf{T}, \mathbf{t}^{\text{sys}}). \tag{D.1}$$

The system tariff remains feasible, so the gain is nonnegative. Competitive equilibrium is recomputed at the deviation profile, and numerical upper bounds remain in place.

Median computed gains are 0.81 log percentage points for advanced economies and 0.39 for developing economies, about 27 and 20 percent of their respective median system values. Singapore, Israel, Mexico, and Switzerland have the largest baseline gains. After climate change, the group medians are 0.81 and 0.41 points. In the United States calculation, no numerical upper bound is reached and all 33 other named economies lose, with the largest losses in Mexico and Canada (Table D3).

Table D3: United States unilateral deviation and partner spillovers

Economy or aggregate	Real-income change
United States	+0.899
Other named economies, income-weighted aggregate	−0.273
Other optimizing governments, median	−0.212
Mexico	−5.124
Canada	−3.280
Israel	−1.020

Notes: Changes are $100 \log(W_i^{\text{dev}}/W_i^{\text{sys}})$, in log percentage points, at baseline productivity. Partners retain their system tariffs. The aggregate uses baseline-income weights renormalized over the 33 named economies other than the United States. The rest-of-world aggregate is excluded. The median covers the 31 other optimizing governments. The final three rows report the largest partner losses.

These gains and spillovers hold partner tariffs fixed and do not establish incentives under retaliation or repeated interaction. Allowing partners to reoptimize under retained restraints would define an asymmetric policy equilibrium. Comparing it with the full system would measure the cost of departure, while comparing it with noncooperation would value the remaining restraints. The bilateral conflict exercise in Section 5 instead imposes only the United States–China tariff pair.

E Conflict tariffs and additional results

This appendix documents the imposed tariff schedules and the additional outcomes and diagnostics supporting the conflict application.

E.1 Product universe and tariff aggregation

The schedules retain the baseline product-sector mapping, covering 4,595 HS6 products imported by the United States and 4,610 imported by China. Excluded products account for 0.605

and 5.111 percent of the respective raw bilateral values. All covered products remain in the denominator when missing tariff inputs are completed.

For importer i, exporter j, sector s, product p, and national line n within that product, the imposed tariff index is

$$q_{ji}^{s,r} = \sum_{p \in s} \phi_{jip}^{0} \sum_{n \in p} \eta_{jipn}^{0} \left(o_{jipn}^{r} + d_{jipn}^{r} - h_{jipn}^{r} \right), \qquad r \in \{0, 1\}. \tag{E.1}$$

The HS6 weights ϕ^0 use fixed raw BACI imports and sum to one within each bilateral sector. The national-line weights η^0 sum to one within each HS6 product. Ordinary rates are o, additional duties are d, and eligible relief is h. These arithmetic indices preserve fixed-weight tariff averages, but not every welfare effect of heterogeneous product tariffs.

United States national-line shares use 2017 bilateral Census values from the replication archive of Fajgelbaum et al. (2024). Chinese shares use the 2017 import-value sheet of the Bown, Jung, and Zhang workbook dated June 12, 2019 in that archive. Chinese HS8 weights are divided equally among finer national lines when their shares are unavailable. Missing United States shares are replaced by equal national-line weights. If any positive Chinese HS8 share cannot be mapped, or the total is unavailable, the entire HS6 product receives equal HS8 weights and equal weights within each HS8. The product records identify these completions.

Raw covered trade supplies the tariff weights. The initial equilibrium uses the balanced flows obtained after accounting reconciliation and deficit removal (Appendix C.2). The respective values are \$489.70 billion and \$312.25 billion for Chinese exports to the United States, and \$131.50 billion and \$210.60 billion for United States exports to China. The calibration is unchanged when the schedules are imposed, and trade subsequently adjusts in equilibrium.

Table E1 reports the resulting sectoral tariff schedules at the two endpoints.

Table E1: Imposed bilateral tariffs

Importer	Sector	Preconflict	Conflict
United States	Agriculture	1.032	21.388
United States	Extraction	0.320	6.261
United States	Manufacturing	2.735	17.799
China	Agriculture	5.029	33.508
China	Extraction	0.407	12.029
China	Manufacturing	6.843	13.889

Notes: Ad valorem tariff rates, expressed in percent, on imports from the other conflict country. The indices use fixed 2017 product weights and the dated duty components documented in this appendix. Each schedule is fixed across policy regimes at the respective endpoint. Table E8 compares the conflict rates with initial mean tariffs on other suppliers.

E.2 Tariff construction and conditional policy domains

Annual WITS MFN ad-valorem-equivalent schedules supply ordinary tariffs for 2017 and 2019. Missing rates receive matched-cell means, with the 2.4 percent United States crude-oil proxy retained. Completions are recorded by product and endpoint in `proposal1_tariff_lines.csv`.

Additional duties and exclusions are dated separately, so the indices combine annual ordinary-rate proxies with endpoint measures rather than exact December 31 ordinary schedules.

Episode records come from Fajgelbaum et al. (2020, 2024), the Bown, Jung, and Zhang workbook, and official notices identified in `proposal1_manifest.json`, which records source URLs and file hashes. A measure enters only if announced by the endpoint and effective on that date. Amendments replace earlier versions of the same action, distinct additions are summed, and relief applies only to its duty family. Later-announced relief and suspended December measures are excluded.

United States Lists 1–3 carry additions of 25 percent and List 4A of 15 percent. Suspended List 4B is excluded. Active whole-code exclusions and documented historical product correspondences are applied. Description-limited exclusions receive no unobserved relief in the principal input and full eligible-code relief in the lower-tariff alternative. Chinese June amendments, September additions, automobile suspensions, and applicable exclusions are incorporated. The pre-existing Chinese sugar safeguard enters both endpoints where applicable.

Section 232 duties and safeguards enter the bilateral indices. The principal safeguard input uses second-year United States rates and assumes covered out-of-quota goods, while alternatives allow relief when eligibility or quota use is unknown. Firm-specific exclusions, domestic VAT, and antidumping and countervailing duties are excluded. Historical Section 232 changes against third countries are not separately imposed in the policy game.

Three HS6 products have unusually high ordinary ad-valorem equivalents. The affected United States manufacturing products account for approximately 0.006 percent of covered sector imports but contribute 0.180 tariff percentage points to the 2017 ordinary index. The corresponding Chinese figures are 0.0003 percent and 0.009 points. Source and conversion checks support their provenance, not their economic accuracy. Independent reconstruction reconciles the six principal tariff indices, and Table E2 reports completion and uncertainty shares.

Table E2: Tariff-input completion and uncertainty shares

Importer	Sector	Unknown composition	Rate or eligibility uncertainty	Missing ordinary rate, 2017	Missing ordinary rate, 2019
China	Agriculture	0.1822	2.1135	0.0000	0.0000
China	Extraction	0.0627	0.0000	0.0000	0.0000
China	Manufacturing	5.3414	0.4229	0.0431	0.0000
United States	Agriculture	0.0930	0.1909	0.0000	0.0302
United States	Extraction	0.0063	0.0000	4.2165	0.0000
United States	Manufacturing	0.1318	13.6661	0.2711	0.1782

Notes: Percent of covered bilateral sector imports from the other conflict country. Unknown composition denotes missing national-line shares. Rate or eligibility uncertainty denotes intervals used in the alternative inputs. Missing ordinary rates include the crude-oil proxy. Flags can overlap and need not imply a change in the sector rate.

After ordinary regime domains are constructed, both bounds on each imposed coordinate are replaced by its specified rate. This follows the adjustment that otherwise raises ceilings to observed rates, allowing the conflict tariff to exceed its original ceiling. The same six coordinates are fixed in the matched preconflict state. All other ceilings, background policies, and numerical bounds retain their regime-specific values.

The full system and cooperation alone retain the complete estimated welfare-weight matrix, including the two United States–China weights. Commitments alone and noncooperation set all foreign weights to zero. Each government optimizes its remaining tariffs under the corresponding objective. Alternative institutional assumptions are reported in Appendix G.3.

E.3 Additional regime and national outcomes

Table E3 reports national values and income changes. The difference between the two income changes equals the change in value before rounding. Third-country aggregates include Saudi Arabia and the United Arab Emirates, whose tariffs remain fixed, and exclude rest of world. Figure 2 instead shows optimizing governments only.

Table E3: Remaining full-system value and conflict income changes by economy

	Full-system value			Conflict-induced income change	
Economy	Preconflict	Conflict	Change	With restraint	Without restraint
ARE	3.1621	3.2975	0.1354	-0.01247	-0.14789
ARG	0.5089	0.5844	0.0756	-0.00013	-0.07569
AUS	2.8784	3.0524	0.1740	-0.03316	-0.20718
BGD	2.7586	2.9010	0.1424	0.04862	-0.09380
BRA	0.8728	0.9628	0.0900	0.00003	-0.08994
CAN	3.4469	4.1059	0.6589	0.09843	-0.56052
CHE	5.5842	5.7738	0.1896	0.00281	-0.18679
CHL	2.8351	3.0791	0.2440	-0.00569	-0.24969
CHN	0.0271	0.0238	-0.0033	-0.11682	-0.11349
COL	1.8256	2.0106	0.1849	0.00893	-0.17599
EU	0.1153	0.2224	0.1070	0.01920	-0.08784
IDN	0.7687	0.8295	0.0607	-0.00200	-0.06273
IND	0.3041	0.3461	0.0420	0.00552	-0.03644
IRN	2.3395	2.4503	0.1108	0.01539	-0.09541
ISR	2.6110	2.9261	0.3151	0.09797	-0.21713
JPN	0.4377	0.5466	0.1089	0.01160	-0.09726
KAZ	3.0614	3.1956	0.1342	0.02053	-0.11365
KOR	1.5980	1.7340	0.1360	-0.00226	-0.13825
MEX	5.6392	6.5576	0.9183	0.20053	-0.71781
MYS	3.9930	4.2450	0.2520	0.03790	-0.21409
NOR	5.4476	5.5551	0.1075	0.00195	-0.10553
PAK	0.9551	1.0230	0.0678	0.01135	-0.05649
PER	1.5212	1.6617	0.1405	-0.00699	-0.14746
PHL	3.8598	4.1687	0.3089	0.02780	-0.28107
RUS	1.4120	1.4702	0.0581	0.00200	-0.05614
SAU	2.3927	2.5279	0.1352	-0.02702	-0.16225
SGP	3.5190	3.7553	0.2363	-0.00970	-0.24597
THA	2.3165	2.4711	0.1546	0.02622	-0.12835
TUR	1.4246	1.4688	0.0442	0.00281	-0.04139
TWN	1.4511	1.6140	0.1629	-0.01077	-0.17368
UKR	2.0067	2.0749	0.0682	0.00738	-0.06085
USA	0.3736	0.2914	-0.0822	-0.16848	-0.08629
VNM	1.6779	1.8360	0.1581	0.00843	-0.14967
ZAF	1.9754	2.0905	0.1152	0.01261	-0.10255

Notes: All income comparisons are in log percentage points. Preconflict and Conflict report the full system's real-income gain over noncooperation under the same imposed bilateral schedule. Change is their difference. The final two columns report conflict-induced income changes relative to each regime's matched preconflict equilibrium. ARE and SAU retain fixed background tariffs. Rest of world is excluded. Sign counts use unrounded values.

Table E4 complements the main third-country decomposition with world and conflict-country

income effects under all four regimes. Under cooperation alone, United States tariff adjustment more than offsets its exposure loss. This differs from a rise in institutional value, which compares income with the noncooperative outcome.

Table E4: World and conflict-country income effects

Population	Regime	Exposure	Adjustment	Total
World	Full system	-0.0544	0.0001	-0.0543
World	Commitments only	-0.0544	-0.0004	-0.0548
World	Cooperation only	-0.0541	-0.0063	-0.0604
World	Noncooperation	-0.0542	-0.0637	-0.1179
United States	Full system	-0.1825	0.0140	-0.1685
United States	Commitments only	-0.1825	0.0005	-0.1820
United States	Cooperation only	-0.1483	0.2409	0.0926
United States	Noncooperation	-0.1257	0.0394	-0.0863
China	Full system	-0.1125	-0.0043	-0.1168
China	Commitments only	-0.1132	-0.0003	-0.1135
China	Cooperation only	-0.1193	0.0057	-0.1136
China	Noncooperation	-0.1274	0.0139	-0.1135

Notes: Entries are changes in log real income multiplied by 100, reported in log percentage points. World entries use fixed baseline-income weights over the 34 named economies. Total compares the conflict equilibrium with the matched preconflict equilibrium under the same regime. Exposure imposes the conflict bilateral tariffs while holding all remaining tariffs at that regime's preconflict rates. Adjustment measures the additional income change when remaining tariffs are reoptimized. Components sum to the total before rounding. Differences from the noncooperative row give changes in institutional value and their components.

Table E5 compares each restraint with conditional noncooperation. The stand-alone values overlap and are not additive. China's cooperation-only change is near zero and varies in sign across accepted starting profiles, from -0.000136 to $+0.000274$ points. The memo reports unrestricted baseline values for comparison, without implying matching policies or national outcomes.

E.4 Tariff responses and marginal welfare effects

Table E6 reports mean tariffs over each conflict country's 99 remaining supplier-sector coordinates. Its unrestricted benchmark also optimizes the six bilateral tariffs, which equal 18.195 percent on Chinese goods and 18.250 percent on United States goods in every sector. All exceed their imposed preconflict rates. Appendix D.2 explains the unrestricted tariff structure.

Two unilateral optimizations isolate the initial change in protection incentives. All other governments retain their matched preconflict noncooperative tariffs, while the six bilateral tariffs take their conflict values. The United States best-response mean rises from 14.89 to 19.27 percent and China's from 16.87 to 18.82 percent. All 99 remaining tariffs rise in each response. The optimizations are independent, and neither response is supplied to the other's problem.

The marginal diagnostic follows the same direction across four imposed-tariff states for each government. Let $d_{i\ell} = \log(1+t_{i\ell}^{\mathrm{BR}}) - \log(1+t_{i\ell}^{0})$ and $t_{i\ell}(\varepsilon) = \exp[\log(1+t_{i\ell}^{0}) + \varepsilon d_{i\ell}] - 1$. Derivatives are evaluated at $\varepsilon = 0$, with other remaining tariffs at their initial noncooperative values. Imposed tariffs stay fixed within each state, while competitive-equilibrium prices, expenditure, and trade

Table E5: Values of the full system and each restraint during the conflict

Population	Restraint	Preconflict	Conflict	Change
World	Full system	0.6718	0.7354	0.0636
	Commitments only	0.6035	0.6666	0.0631
	Cooperation only	0.6517	0.7092	0.0575
Third countries	Full system	1.1851	1.3300	0.1449
	Commitments only	1.0702	1.2152	0.1450
	Cooperation only	0.8982	0.9656	0.0674
United States	Full system	0.3736	0.2914	-0.0822
	Commitments only	0.3337	0.2380	-0.0957
	Cooperation only	0.7186	0.8975	0.1789
China	Full system	0.0271	0.0238	-0.0033
	Commitments only	0.01700	0.01697	-0.00003
	Cooperation only	0.2674	0.2673	-0.0001
Memo: Unrestricted baseline full-system values				
World: 0.8030		Third countries: 1.3235		

Notes: Preconflict and Conflict report real-income gains over the indicated noncooperative benchmark, in log percentage points. Change is the conflict value minus its preconflict value. The principal rows compare each restraint with noncooperation under the same imposed bilateral schedule. Memo values compare the full system with unrestricted noncooperation, allowing the six bilateral tariffs to differ across regimes. Stand-alone commitments and cooperation values are overlapping comparisons and are not additive.

Table E6: Mean tariffs on remaining suppliers

Importer	Regime	Preconflict	Conflict
A. Matched bilateral schedules			
United States	Full system	1.3291	1.4083
	Commitments only	1.6256	1.6526
	Cooperation only	7.6645	9.9976
	Noncooperation	14.8908	19.3176
China	Full system	2.8004	2.7647
	Commitments only	3.9073	3.9147
	Cooperation only	8.2703	8.5110
	Noncooperation	16.8717	18.9859
B. Unrestricted baseline			
Importer	Regime	Mean tariff	
United States	Noncooperation	19.3588	
China	Noncooperation	19.4796	

Notes: Baseline-import-weighted means of ad valorem tariff rates, expressed in percent, over the 99 remaining supplier-sector coordinates of each importer, including rest of world and zero-rate cells. The conflict partner is excluded. For each importer, support and weights are fixed across regimes and states. Panel B allows the six bilateral tariffs to be chosen endogenously.

respond.

Let a dot denote differentiation with respect to ε and write $u_j = d\log w_j/d\varepsilon$. With fixed technology and iceberg costs, balanced trade implies

$$I_i \frac{d\log W_i}{d\varepsilon} = \sum_{j\neq i,s} \left(M^s_{ij}u_i - M^s_{ji}u_j\right) + \sum_{j\neq i,s} t^s_{ji}\left(\dot{M}^s_{ji} - M^s_{ji}u_j\right). \tag{E.2}$$

The first term is the terms-of-trade contribution. The second values changes in distorted import

quantities at current producer prices. Table E7 divides the latter between the conflict partner and other suppliers, including rest of world. Partner imports respond despite their fixed tariff. The direct tariff-revenue change cancels the direct consumer-price effect.

Table E7: Marginal welfare effects of raising remaining tariffs

Government	Imposed direction	Total	Terms of trade	Partner distortion	Other-supplier distortion
United States	Neither	3.51×10^{-7}	0.227113	0.011736	-0.238849
	United States	0.072208	0.253800	0.046013	-0.227605
	China	0.002985	0.219639	0.011446	-0.228100
	Both	0.072859	0.245280	0.044683	-0.217104
China	Neither	-2.85×10^{-9}	0.053478	0.003529	-0.057007
	United States	-0.001223	0.049418	0.003357	-0.053998
	China	0.009093	0.057157	0.006312	-0.054377
	Both	0.006878	0.052696	0.005776	-0.051594

Notes: Entries are $100\, d \log W_i / d\varepsilon$, measured in log percentage points of real income per unit increase in the path parameter ε and evaluated at $\varepsilon = 0$. For each government, $\varepsilon = 0$ gives the initial remaining tariffs and $\varepsilon = 1$ gives its saved unilateral best response, using the log-gross-tariff path defined above. Magnitudes are comparable across states within a government and are not finite adjustment gains. An imposed direction names the government raising its bilateral tariffs. All other remaining tariffs stay fixed. Components use current-state household income and sum to the total before rounding.

The conflict reduces the net marginal distortion cost for both governments, while that cost remains positive. The terms-of-trade contribution increases for the United States but decreases for China. Each government's own bilateral barrier raises its incentive along the evaluated direction, whereas the United States barrier alone lowers China's incentive. The separate directions interact and need not sum to the joint effect. Analytic components satisfy finite-difference and adding-up checks at a tolerance of 10^{-8} in natural-log units.

E.5 National decompositions and cooperation alone

Panel A of Table E9 separates changes in value into exposure and own- and partner-policy contributions. Partner-policy differences account for almost all of the third-country increase. United States exposure and own-policy adjustment lower its value, partly offset by partner policies. China's positive exposure difference is outweighed by its adjustment difference. National exposure differences nearly cancel in the world aggregate, whose fixed weights reflect the calibrated tradable economy rather than world GDP.

The exposure calculation also separates the two imposed tariff directions. Let e_i^U and e_i^C be income changes from imposing only the United States or Chinese direction at initial remaining tariffs, and let e_i^{UC} impose both. The interaction is $h_i = e_i^{UC} - e_i^U - e_i^C$. Symmetric contributions are $e_i^U + h_i/2$ and $e_i^C + h_i/2$. Panel B subtracts each noncooperative component from its full-system counterpart. These are contributions of tariff directions, not separate import and export channels.

Table E8 describes the initial tariff comparisons. At initial system policies, conflict rates exceed mean tariffs on other suppliers in every sector. At initial noncooperative policies, they

reverse the bilateral tariff advantage in agriculture for both countries and in United States manufacturing. The advantage narrows but remains elsewhere. These rankings describe the policy changes but do not identify their welfare contributions.

Table E8: Conflict tariffs and initial protection against other suppliers

		Imposed	Mean on other suppliers	
Importer	Sector	conflict rate	Full system	Noncooperation
United States	Agriculture	21.388	0.431	16.195
United States	Extraction	6.261	1.503	15.739
United States	Manufacturing	17.799	1.340	14.780
China	Agriculture	33.508	4.562	16.177
China	Extraction	12.029	0.283	16.761
China	Manufacturing	13.889	3.626	16.960

Notes: Ad valorem tariff rates, expressed in percent. The imposed conflict rate applies to imports from the other conflict country. Means on other suppliers use fixed calibrated-import weights within each importer-sector cell, including zero-rate cells and rest of world, at each regime's matched preconflict tariffs.

Panel C of Table E9 reverses the sequence by evaluating adjustment in the initial environment and exposure at final tariffs. Adjustment accounts for approximately 98 percent of the third-country increase under the principal order and 96 percent under the reverse order. The United States adjustment component changes sign. These allocations preserve total value changes but do not identify unique causal shares.

Under cooperation alone, both own and partner policies contribute to the United States adjustment gain (Table E10). Partner policies are evaluated first and include tariffs on every exporter. A cooperative choice need not make domestic income locally stationary with respect to its own tariffs, unlike a smooth interior noncooperative optimum. This local distinction does not determine the finite gain.

The mean tariff on United States goods imposed by the 30 optimizing third-country governments falls from 4.79 to 3.93 percent under cooperation alone, with all 90 tariffs declining. Under noncooperation the mean barely changes, with 22 declines and 68 increases. Means use fixed calibrated-import weights over the same coordinates. These tariffs are only part of the partner response and do not separately account for the broad partner-policy contribution.

A local diagnostic evaluates cooperative gradients at initial remaining tariffs under the two bilateral schedules. For a tariff ℓ chosen by government k on United States goods, the domestic, United States, and other-foreign components are

$$\frac{1}{I_k^0}\frac{\partial G_k}{\partial t_\ell} = D_{k\ell} + U_{k\ell} + O_{k\ell}, \tag{E.3}$$

$$D_{k\ell} = \widehat{W}_k \frac{\partial \log \widehat{W}_k}{\partial t_\ell},$$

$$U_{k\ell} = \lambda_{k,\mathrm{US}} \frac{I_{\mathrm{US}}^0 \widehat{W}_{\mathrm{US}}}{I_k^0} \frac{\partial \log \widehat{W}_{\mathrm{US}}}{\partial t_\ell},$$

$$O_{k\ell} = \sum_{j \notin \{k,\mathrm{US}\}} \lambda_{kj} \frac{I_j^0 \widehat{W}_j}{I_k^0} \frac{\partial \log \widehat{W}_j}{\partial t_\ell}.$$

Table E9: Decompositions of conflict-induced changes in full-system value

Panel A. Exposure and policy contributions

Population	Exposure	Own policy	Partner policy	Total
World	−0.000152	−0.005283	+0.069066	+0.063631
Third countries	+0.003367	+0.000294	+0.141251	+0.144912
United States	−0.056806	−0.029008	+0.003624	−0.082190
China	+0.014866	−0.004994	−0.013203	−0.003331

Panel B. Tariff-direction contributions to the exposure difference

	Separate changes and interaction			Symmetric contributions		
Population	US	China	Interaction	US	China	Total
World	+0.000219	−0.003189	+0.002818	+0.001628	−0.001780	−0.000152
Third countries	+0.003812	−0.004749	+0.004304	+0.005964	−0.002597	+0.003367
United States	−0.178228	+0.126687	−0.005264	−0.180860	+0.124055	−0.056806
China	+0.058478	−0.047134	+0.003521	+0.060239	−0.045373	+0.014866

Panel C. Attribution under alternative calculation orders

Population	Calculation order	Exposure	Adjustment	Total
United States	Exposure first	−0.056806	−0.025384	−0.082190
	Adjustment first	−0.101803	+0.019613	−0.082190
	Symmetric average	−0.079304	−0.002885	−0.082190
Third countries	Exposure first	+0.003367	+0.141545	+0.144912
	Adjustment first	+0.005888	+0.139023	+0.144912
	Symmetric average	+0.004628	+0.140284	+0.144912

Notes: Entries are in log percentage points. World and third-country aggregates use fixed baseline-income weights over 34 and 32 named economies, respectively. Panel A decomposes the conflict-induced change in the full system's real-income gain over noncooperation. Exposure is evaluated first, and policy components use the partners-first ordering in Appendix B.3. Panel B decomposes only the exposure difference between regimes. Directions identify the tariff-imposing country. Separate changes begin at each regime's preconflict policies, and each symmetric contribution includes half the interaction. Panel C decomposes the United States and third-country value changes. Exposure first evaluates the imposed tariff change at initial remaining policies. Adjustment first evaluates remaining tariff changes in the initial environment, then exposure at final policies. The symmetric row averages the two allocations. Components sum to their respective totals before rounding.

Table E10: United States adjustment gains under cooperation and noncooperation

Regime	Partner policies	Own policies	Total
Cooperation only	+0.1099	+0.1310	+0.2409
Noncooperation	+0.0035	+0.0359	+0.0394
Cooperation minus noncooperation	+0.1064	+0.0951	+0.2015

Notes: The first two rows report United States real-income gains from adjusting remaining tariffs, in log percentage points. The imposed conflict schedule is fixed, and adjustment begins from each regime's matched preconflict remaining tariffs. The final row subtracts noncooperation from cooperation alone. All partners' remaining tariffs change before United States remaining tariffs. The partner block includes tariffs on every exporter. Components sum before rounding.

The components sum to the complete objective gradient divided by I_k^0. Derivatives use decimal tariff rates, and the level factors preserve the baseline-price objective. A feasible immediate cut is supported when $[(1+t_\ell)/G_k](\partial G_k/\partial t_\ell) < -10^{-6}$.

The United States component becomes more negative at all 90 coordinates, while the other-foreign component moves in the opposite direction. The complete gradient supports an immediate cut at 32 coordinates covering 94.48 percent of relevant baseline imports. No coordinate supports an increase beyond the tolerance, and the other 58 establish neither direction. Subsequent simultaneous adjustment reduces all 90 tariffs. United States receipts from these markets fall by 1.17 percent at fixed remaining cooperative tariffs, so larger marginal foreign-income losses do not arise from an increase in aggregate sales to them.

F Climate inputs, results, and reforms

This appendix documents the productivity inputs, national climate outcomes, and reforms of cooperation and commitments.

F.1 Climate inputs and mappings

The productivity projections are assembled by Nath (2025) for 2080 to 2099 under RCP 8.5. For economies with national agricultural inputs, the principal multiplier averages available projections from the Climate Impact Lab (Hultgren et al., 2025), Cline (2007) including carbon fertilization, and Iglesias and Rosenzweig (2009). It excludes the crop-model input of Costinot et al. (2016). The European Union and rest of world retain the prepared regional aggregates rather than an independent reconstruction from their source components. Table F1 reports the inputs and proxy assignments.

Agricultural-source sensitivities replace inputs where available and retain the principal multiplier elsewhere. Manufacturing uses one plus the projected proportional change in revenue per worker and is unchanged across these sensitivities. Extraction is unshocked. Productivity enters as $\widehat{T}_{is} = (\widehat{a}_{is})^{\theta_s}$, without re-estimation. The common-scale neutrality result is established in Appendix A.4.

F.2 National and sectoral climate results

Table F2 reports national income changes, institutional values, and the exposure and adjustment contributions. In the world aggregate, adjustment adds 0.0046 log percentage points to value. Evaluating partners' changes first assigns 0.0042 points to partner policies and 0.0004 to own policies. These are differences in adjustment across the full system and noncooperation under the ordering of Appendix B.3.

Sector-component experiments show that national incidence also depends on foreign productivity changes. For Australia, agricultural changes alone reduce value by 0.019 points, manufacturing changes alone by 0.241 points, and the combined shock by 0.263 points. Its own

Table F1: The climate shock by economy

Economy	Agriculture	Manufacturing	Economy	Agriculture	Manufacturing
ARE	−19.2	0.2	MEX	−16.5	0.0
ARG	−9.8	0.0	MYS	−9.5	0.0
AUS	−24.2	0.0	NOR	11.1	−0.6
BGD	−4.7	−1.6	PAK	−22.1	0.1
BRA	−4.5	0.0	PER	−16.2	−0.5
CAN	−6.2	−0.7	PHL	−9.6	0.0
CHE	−4.5	−3.1	RUS	−17.0	−0.6
CHL	−10.9	−0.9	SAU	−19.2	0.2
CHN	−6.2	−3.6	SGP	−13.5	0.0
COL	−13.0	0.0	THA	−17.5	0.0
EU	−9.0	−1.3	TUR	−14.9	−3.8
IDN	−9.8	0.0	TWN	−6.2	−3.6
IND	−17.5	0.0	UKR	−13.7	−3.8
IRN	−19.2	0.2	USA	−12.4	0.3
ISR	−24.0	0.0	VNM	−6.3	0.0
JPN	0.4	−1.0	ZAF	−23.1	0.0
KAZ	−0.7	−3.1	RoW	−19.1	−2.7
KOR	19.9	−1.5			

Notes: Entries are percentage changes in productivity, $100(\widehat{a}_{is} - 1)$, for the principal climate scenario. Extraction is unchanged. Economies are ordered by code, with rest of world last. Taiwan uses China's input, Saudi Arabia and the United Arab Emirates use Iran's, and the European Union and rest of world retain the supplied regional aggregates. Entries rounded to zero need not be exactly zero.

manufacturing productivity changes by less than 0.0001 percent. These experiments change sectoral productivity across all economies and are not an additive decomposition of national channels.

Retaining commitments in one sector only, with zero foreign weights, gives a baseline value of 0.076 log percentage points for agriculture, 0.220 for extraction, and 0.724 for manufacturing, compared with 0.731 for all commitments. These values are not additive. Manufacturing commitments have the largest stand-alone value, while projected manufacturing productivity changes are small. This diagnostic concerns existing ceilings and does not bound the gains from deeper agricultural commitments.

F.3 Agricultural standstill

The reform prevents already covered agricultural tariffs from rising above their solved baseline system rates. Let $\mathcal{A}$ contain the 972 covered agricultural instruments chosen by optimizing governments and let $t_k^{\text{sys},0}$ be instrument k's baseline system rate. The reformed ceiling is

$$\bar{t}_k^{\text{tight}} = \begin{cases} \min\{\bar{t}_k, t_k^{\text{sys},0}\}, & k \in \mathcal{A}, \\ \bar{t}_k, & k \notin \mathcal{A}. \end{cases}$$

The same vector applies at baseline and climate productivity. All other bounds, uncovered choices, fixed policies, and welfare weights remain unchanged. The final domains are checked against these restrictions when solving and verifying the reform.

Restricting a feasible set while retaining an original best response preserves that response.

Table F2: Income and institutional value under climate change

	Climate income change			Full-system value			Change decomposition	
Economy	Observed tariffs fixed	System	Noncoop.	Baseline	Climate	Change	Exposure	Adjustment
ARE	−0.620	−0.548	−0.508	3.319	3.279	−0.040	−0.089	0.049
ARG	−1.281	−1.404	−1.396	0.590	0.582	−0.008	0.036	−0.044
AUS	−1.970	−2.082	−1.819	3.094	2.831	−0.263	−0.246	−0.016
BGD	−2.273	−2.343	−2.327	2.887	2.871	−0.016	−0.017	0.001
BRA	−0.675	−0.618	−0.567	0.969	0.918	−0.051	−0.075	0.024
CAN	−0.691	−0.560	−0.635	4.015	4.090	0.075	0.066	0.009
CHE	−2.856	−2.867	−2.881	5.768	5.782	0.014	−0.003	0.016
CHL	−1.745	−1.725	−1.629	3.100	3.004	−0.096	−0.105	0.009
CHN	−3.659	−3.656	−3.674	0.133	0.151	0.019	0.017	0.002
COL	−2.850	−2.838	−2.736	2.012	1.909	−0.102	−0.107	0.005
EU	−1.617	−1.608	−1.619	0.213	0.224	0.011	0.004	0.008
IDN	−1.878	−1.851	−1.795	0.840	0.783	−0.057	−0.059	0.002
IND	−4.348	−4.282	−4.277	0.346	0.342	−0.004	−0.002	−0.003
IRN	−3.763	−3.805	−3.723	2.459	2.378	−0.082	−0.065	−0.016
ISR	−1.218	−1.087	−1.111	2.852	2.877	0.025	−0.023	0.048
JPN	−0.982	−1.049	−1.035	0.544	0.530	−0.014	−0.021	0.007
KAZ	−1.594	−1.558	−1.612	3.195	3.249	0.054	0.093	−0.040
KOR	−1.082	−1.625	−1.635	1.743	1.753	0.010	0.002	0.008
MEX	−1.676	−1.659	−1.637	6.369	6.346	−0.022	−0.052	0.030
MYS	−1.142	−1.136	−1.077	4.234	4.175	−0.059	−0.084	0.025
NOR	0.135	0.088	0.193	5.562	5.457	−0.105	0.024	−0.129
PAK	−7.174	−7.191	−7.163	1.026	0.997	−0.028	0.015	−0.043
PER	−2.342	−2.314	−2.209	1.677	1.571	−0.105	−0.109	0.004
PHL	−2.304	−2.213	−2.139	4.170	4.096	−0.074	−0.077	0.003
RUS	−2.130	−2.195	−2.136	1.476	1.417	−0.059	−0.049	−0.009
SAU	−1.246	−1.266	−1.161	2.562	2.457	−0.105	−0.101	−0.004
SGP	−0.206	−0.210	−0.235	3.770	3.794	0.025	−0.009	0.034
THA	−1.352	−1.271	−1.316	2.464	2.510	0.046	0.010	0.035
TUR	−5.155	−5.353	−5.423	1.471	1.541	0.070	0.098	−0.028
TWN	−3.462	−3.441	−3.486	1.628	1.673	0.045	0.028	0.018
UKR	−5.436	−5.167	−5.263	2.077	2.174	0.096	0.120	−0.024
USA	−0.479	−0.468	−0.436	0.548	0.516	−0.032	−0.042	0.010
VNM	−0.732	−0.677	−0.652	1.855	1.830	−0.026	−0.025	0.000
ZAF	−2.341	−2.961	−2.845	2.092	1.977	−0.116	−0.140	0.024
World	−2.307	−2.323	−2.320	0.803	0.800	−0.0034	−0.0080	0.0046

Notes: All entries are in log percentage points. Income changes compare climate with baseline productivity at observed tariffs held fixed or at the indicated regime's tariffs, solved at each state. Full-system value is its real-income gain over noncooperation at the same productivity. Change is the climate value minus the baseline value. Exposure changes productivity at each regime's own baseline tariffs. Adjustment adds the difference in subsequent tariff responses across regimes. The world row uses fixed baseline-income weights over the 34 named economies. Saudi Arabia and the United Arab Emirates retain fixed tariffs. Entries are rounded independently from unrounded calculations.

The retained baseline profile passes the tightened-domain checks and leaves baseline welfare unchanged to numerical precision, subject to the numerical qualification in Appendix B.4. The original full-system tariff profile at climate productivity violates 318 tightened ceilings on flows representing 0.826 percent of baseline modeled cross-border imports. A new climate policy equilibrium is therefore required. Table F3 reports the domain diagnostics and welfare effects.

For economy i, the climate-state reform gain is

$$m_i = 100 \left[\log W_i^{\text{tight}}(\mathbf{T}^1) - \log W_i^{\text{sys}}(\mathbf{T}^1) \right].$$

This gain, in log percentage points, also measures additional mitigation because the baseline system outcome is preserved. Noncooperation is unchanged, so it also equals the reform's effect on the climate-induced change in institutional value.

Table F3: Agricultural standstill and welfare effects

Panel A. Domain diagnostics	
Targeted agricultural instruments	972
Original climate choices above the new ceilings	318
Their share of all baseline cross-border imports (percent)	0.826
Their share of targeted agricultural imports (percent)	26.70
Maximum ceiling exceedance (tariff percentage points)	0.721
Panel B. Climate-state real-income gains, log percentage points	
World	0.000106
Median optimizing government	0.000068
Ukraine, largest national gain	0.04923
India, largest national loss	−0.00265

Notes: Panel A evaluates ceiling violations at the original climate system profile. Import shares use fixed baseline flows and the indicated denominators, counting linked bilateral flows once. Panel B compares the reformed and original full systems at climate productivity. World uses fixed baseline-income weights over the 34 named economies, national extrema cover those economies, and the median covers 32 optimizing governments. All other institutional parameters remain fixed. The baseline reform gain is zero to numerical precision, so the climate-state gain also measures additional mitigation.

The world gain offsets approximately 0.005 percent of the original system's climate income loss. Aggregate mitigation is therefore negligible despite constraints on some tariff increases and differences across national outcomes. Numerical tolerances do not bound welfare error, so the interpretation rests on the small magnitude rather than its positive sign. The result concerns this standstill under the full system, not other agricultural instruments or a commitments-only reform.

F.4 Broader institutional reforms

Deeper agricultural commitments and stronger cooperation provide negligible additional aggregate mitigation in the five designs considered here. A reform can raise income both with and without climate change, so its contribution to mitigation requires a comparison across productivity states.

Five permanent reforms are solved at both states, retaining all other primitives and fixed policies. Two cap all 1,009 eligible agricultural instruments at 5 percent or zero, extending beyond the standstill's already covered subset. The 5 percent cap is the minimum of the existing upper bound and 5 percent, preserving tighter ceilings. Two raise foreign welfare weights in optimizing governments' objectives to 0.75 or one, with ceilings retained. The fifth combines zero agricultural tariffs and a weight of 0.75. Weight changes retain all foreign recipients, including rest of world, and represent stronger internalization rather than estimated effects of particular provisions.

Let $g^r(s)$ be the fixed-weight real-income gain of reform r over the existing system in state s, in log percentage points. Additional mitigation is $M^r = g^r(C) - g^r(0)$, the climate-state

gain net of the baseline gain. Table F4 reports these comparisons for the 34 named economies and two fixed groups. The ten optimizing governments with the largest proportional climate income losses under the existing system are China, Colombia, India, Iran, Pakistan, South Africa, Switzerland, Taiwan, Turkey, and Ukraine. The agricultural-exposure group contains Australia, India, Iran, Israel, Mexico, Pakistan, Peru, Russia, Saudi Arabia, South Africa, Thailand, and the United Arab Emirates, whose agricultural productivity falls by more than 15 percent. Both groups are selected under the principal inputs and retained across reforms.

Table F4: Reforms of the restraints and the climate income loss

Panel A. World

Reform	Baseline gain	Climate-state gain	Mitigation	Percent of loss
Agricultural standstill	0.0000	0.0001	0.0001	0.005
Agricultural cap of 5 percent	0.0008	0.0009	0.0002	0.007
Zero agricultural tariffs	0.0016	0.0030	0.0015	0.063
Foreign weight 0.75	0.0269	0.0272	0.0003	0.012
Foreign weight 1	0.0247	0.0237	−0.0010	−0.043
Zero agricultural tariffs and weight 0.75	0.0240	0.0258	0.0018	0.077

Panel B. Exposed groups

	Ten largest losers		Agricultural exposure	
Reform	Climate-state gain	Mitigation	Climate-state gain	Mitigation
Agricultural standstill	0.0000	0.0000	−0.0008	−0.0008
Agricultural cap of 5 percent	−0.0015	−0.0006	−0.0136	−0.0076
Zero agricultural tariffs	−0.0244	0.0024	−0.0690	−0.0584
Foreign weight 0.75	−0.0148	−0.0063	0.0010	−0.0189
Foreign weight 1	−0.0441	−0.0039	−0.0757	−0.0326
Zero agricultural tariffs and weight 0.75	−0.0357	−0.0025	−0.0575	−0.0559

Notes: Gains compare the reformed and existing full systems at the same productivity, in log percentage points. Mitigation is the climate-state gain less the baseline gain. Aggregates use fixed baseline-income weights within each group. Percent of loss is $100M^r/|L^B|$, where the existing system's world climate income change is $L^B = -2.323$ points. Groups are fixed across reforms and defined in the text. Differences and percentages use unrounded results. The standstill is defined in Appendix F.3.

The largest additional mitigation, from the combined reform, is 0.0018 log percentage points, or 0.08 percent of the climate loss. Raising the foreign weight to 0.75 increases climate-state income by 0.027 points, almost all of which is present without climate change. A weight of one yields a smaller reported gain and slightly increases the proportional climate loss. The reporting index averages named economies' log incomes, whereas government objectives use income levels and include rest of world, so their rankings need not coincide. All solutions pass the acceptance checks of Appendix B.4.

The weighted-average climate-state income of the ten largest losers falls under each of the five reforms. The agricultural-exposure group receives negative additional mitigation under all five. Market access and national income need not move together. Zero agricultural tariffs raise Pakistan's agricultural purchasing power, nominal income divided by the agricultural price index, by 0.25 percent. Its tariff-revenue share falls by 0.48 percentage points and its real income

by 0.26 log percentage points. Ukraine instead gains 0.84 points. Import tariffs redistribute income through prices and revenue, but the framework contains no international compensation instrument.

These results concern the five specified designs. They do not evaluate targeted preferences for exposed exporters or welfare weights that rise with projected damage. International transfers, export restrictions, and adaptation investment lie outside the model.

G Sensitivity

This appendix reports the sensitivity of institutional values to welfare weights, policy domains, conflict inputs, and calibration choices.

G.1 Welfare weights

Table G1 compares the full-system value with the value of each restraint alone at alternative common weights. The confidence endpoints use importer-clustered inference, while the illustrative weight of 0.26 lies below that interval. Each share uses the full-system value at the same weight as its denominator. Cooperation alone can exceed the full system, so these comparisons do not impose a ranking of the regimes.

G.2 Policy domains and estimation

Table G2 crosses the policy domain with the welfare weight. It distinguishes removing preferential ceilings at a fixed weight from re-estimating preferences after that change. The latter changes both the constraints represented in the model and the estimating sample. Its higher estimate therefore does not identify the restraint supplied by preferential commitments alone.

The quantitative model uses independent bilateral-sector choices. In the linked importer-sector tariff sensitivity, nonpreferential tariffs within an importer-sector move by a common additive shift. If bilateral rates are $\widetilde{\mathbf{t}}_i = \mathbf{B}_i\mathbf{x}_i + \mathbf{c}_i$, the relevant policy gradient is $\mathbf{B}_i^{\mathsf{T}}\nabla_{\widetilde{\mathbf{t}}_i} G_i$. This sensitivity aggregates the marginal effects of the linked bilateral choices while preserving their observed rate differences.

The numerical-cap rows in Table G5 change both the policy domain and ceiling construction. Changing either alone at a fixed weight moves world value by less than 0.001 log percentage points at either productivity state. The Gulf extension makes Saudi Arabia and the United Arab Emirates tariff setters using known-binding imputation and the cap where bindings are unavailable, while retaining observed preferential ceilings. It does not treat missing schedules as newly observed.

G.3 Conflict sensitivities

The lower- and higher-tariff inputs vary assumptions about rates, eligibility, and composition. Each uses the same national-line allocation at both endpoints when composition is uncertain. The

Table G1: The value of each restraint alone under alternative weights

Weight specification	λ	Full-system value	Relative to full-system value (%)	
			Cooperation alone	Commitments alone
Illustrative lower weight	0.260	0.769	68.9	95.1
Lower confidence endpoint	0.371	0.785	84.8	93.1
IV estimate using the free-trade instrument	0.408	0.791	88.7	92.5
Baseline estimate	0.500	0.803	96.4	91.0
Upper confidence endpoint	0.628	0.818	102.3	89.4

Notes: Baseline productivity and baseline ceilings. Welfare weights λ are dimensionless. Full-system value is the full system's real-income gain over noncooperation, in log percentage points, and the last two columns report the value under each restraint alone as a percentage of it at the same weight. The confidence endpoints are the importer-clustered 95 percent interval, and the illustrative weight of 0.26 lies below it.

Table G2: Preferential ceilings and the estimated weight

Weight specification	λ	Preferential ceilings retained		Preferential ceilings removed	
		Baseline	Climate	Baseline	Climate
Baseline estimate	0.500	0.803	0.800	0.773	0.769
Re-estimated without preferential ceilings	0.880	0.832	0.831	0.829	0.828

Notes: The welfare weight λ is dimensionless. Baseline and Climate report the full system's real-income gain over noncooperation at the corresponding productivity, in log percentage points. Removing the preferential ceilings retains the WTO bound rates. The re-estimated weight uses the interior sample after preferential ceilings are removed.

ordinary-rates-fixed variant changes additions and relief while retaining 2017 ordinary components. Welfare-weight checks use the corresponding matched preconflict baseline. Table G3 reports the resulting value changes. The tariff alternatives are neither statistical confidence intervals nor welfare bounds.

Table G3: Sensitivity of conflict-induced changes in full-system value

Specification	World	Third countries	United States	China
Principal	0.0636	0.1449	-0.0822	-0.0033
Lower welfare weight	0.0632	0.1441	-0.0881	-0.0013
Upper welfare weight	0.0634	0.1476	-0.0711	-0.0118
Lower-tariff assumptions	0.0568	0.1293	-0.0428	-0.0136
Higher-tariff assumptions	0.0638	0.1452	-0.0812	-0.0036
Ordinary rates fixed	0.0659	0.1488	-0.0683	-0.0076
Zero bilateral weights	0.0628	0.1411	-0.0806	-0.0008
WTO bindings only	0.0642	0.1457	-0.0664	-0.0085

Notes: Entries are conflict-induced changes in the full system's real-income gain over noncooperation, in log percentage points. Each row compares the conflict with its own matched preconflict baseline. Weight endpoints are 0.371 and 0.628. Tariff-input alternatives vary the documented construction assumptions and do not bound welfare. The final two rows retain the principal weight except for the two entries set to zero in the bilateral-weight exercise.

The bilateral-weight sensitivity sets only the United States government's weight on Chinese income and China's weight on United States income to zero. Other weights remain unchanged. The third-country full-system increase falls by approximately 3 percent, while its cooperation-only counterpart falls by approximately three quarters. Commitments-only and noncooperative

outcomes are unchanged by construction.

The WTO-only sensitivity reconstructs ceilings without preferential constraints, including preferential components within rest-of-world suppliers. It retains the estimated weight, applied-rate feasibility adjustments, numerical bounds, and fixed-policy economies. Of the 383 preferential zero singleton domains, 335 become adjustable. Japan's agricultural ceiling on rest-of-world imports falls by approximately 0.009 tariff percentage points, while every other ceiling is unchanged or higher. The exercise therefore compares two source-based domains rather than weakly relaxing every interval.

Table G4 reports values under these alternatives. WTO-only ceilings lower the initial third-country full-system value but leave its conflict-induced increase close to the principal result. China's commitments-only value changes from −0.0301 to −0.0335 points, a decline of 0.0034 points. Cooperation-only and noncooperative outcomes are unaffected by the ceiling change. All reported games satisfy the acceptance criteria in Appendix B.4.

Table G4: Institutional sensitivities with matched baselines

A. Zero bilateral weights				
Population	Restraint	Preconflict	Conflict	Change
World	Full system	0.6716	0.7344	0.0628
	Commitments only	0.6035	0.6666	0.0631
	Cooperation only	0.6538	0.7083	0.0545
Third countries	Full system	1.1850	1.3261	0.1411
	Commitments only	1.0702	1.2152	0.1450
	Cooperation only	0.9576	0.9740	0.0164
United States	Full system	0.3723	0.2918	-0.0806
	Commitments only	0.3337	0.2380	-0.0957
	Cooperation only	0.5901	0.8379	0.2478
China	Full system	0.0274	0.0266	-0.0008
	Commitments only	0.01700	0.01697	-0.00003
	Cooperation only	0.2321	0.2736	0.0415
B. WTO bindings only				
Population	Restraint	Preconflict	Conflict	Change
World	Full system	0.6426	0.7068	0.0642
	Commitments only	0.4664	0.5295	0.0632
	Cooperation only	0.6517	0.7092	0.0575
Third countries	Full system	1.1088	1.2546	0.1457
	Commitments only	0.8776	1.0227	0.1450
	Cooperation only	0.8982	0.9656	0.0674
United States	Full system	0.3927	0.3263	-0.0664
	Commitments only	0.1715	0.0853	-0.0862
	Cooperation only	0.7186	0.8975	0.1789
China	Full system	0.0496	0.0411	-0.0085
	Commitments only	-0.0301	-0.0335	-0.0034
	Cooperation only	0.2674	0.2673	-0.0001

Notes: Preconflict and Conflict report real-income gains over noncooperation under the same imposed bilateral schedule, in log percentage points. Change is their difference. Aggregates use the principal fixed baseline-income weights over 34 named economies for the world and 32 for third countries. Panel A changes only the two bilateral welfare weights. Panel B retains the principal weight and reconstructs ceilings without preferential constraints, including the Japan–rest-of-world exception described in the text. Stand-alone values are not additive.

Calculation order affects attribution rather than the total value change. Appendix Table E9,

Table G5: The value of the system under alternative specifications

	Weight	World value	Climate response	
Specification	λ	Baseline	World	Median
Baseline	0.500	0.803	−0.0034	−0.019
Estimation and welfare weights				
Lower confidence endpoint	0.371	0.785	−0.0035	−0.013
Upper confidence endpoint	0.628	0.818	−0.0034	−0.020
Interior coverage at least one half	0.211	0.762	−0.0035	−0.011
Lower confidence endpoint of the one-half coverage weight	0.020	0.734	−0.0028	−0.001
Government-specific weights	$\{\widehat{\lambda}_i\}$	0.795	−0.0033	−0.013
Margins not scaled	0.658	0.821	−0.0034	−0.019
Margins scaled by cell imports	0.504	0.803	−0.0034	−0.020
Iran included in the estimation sample	0.430	0.794	−0.0034	−0.015
Policy domains and calibration				
Linked importer-sector tariff changes	0.519	0.806	−0.0069	−0.017
Preferential ceilings removed	0.880	0.829	−0.0008	−0.029
Interior classification margin: 2 points	0.499	0.803	−0.0034	−0.019
Interior classification margin: 10 points	0.486	0.801	−0.0034	−0.019
Domain and construction cap: 150 percent	0.500	0.803	−0.0034	−0.019
Domain and construction cap: 300 percent	0.500	0.802	−0.0034	−0.019
Trade elasticities multiplied by 0.8	0.587	0.988	−0.0027	−0.019
Trade elasticities multiplied by 1.2	0.414	0.677	−0.0041	−0.022
Saudi Arabia and UAE as tariff setters	0.490	0.810	−0.0027	−0.023
Climate components and sources, weight fixed				
Agricultural component alone	0.500	0.803	0.0005	−0.004
Manufacturing component alone	0.500	0.803	−0.0045	−0.021
Climate Impact Lab agricultural input	0.500	0.803	−0.0027	0.003
Cline agricultural input	0.500	0.803	0.0123	0.012
Iglesias–Rosenzweig agricultural input	0.500	0.803	0.0003	−0.021
Taiwan assigned South Korea's climate inputs	0.500	0.803	−0.0032	−0.020

Notes: Welfare weights λ are dimensionless. Baseline is the full system's world real-income gain over noncooperation, in log percentage points. Climate response is the change in that value, in the same units. Medians cover the 32 baseline optimizing governments, or all 34 named economies in the Gulf extension. World weights remain the fixed baseline-income shares of the 34 named economies. Classification margins are tariff percentage points, and numerical caps are ad valorem rates in percent. Weights are re-estimated when the estimating sample or margins change, including the elasticity exercises. Endpoint, government-specific, and climate-source rows use the indicated weights. Source alternatives retain the principal agricultural input where the selected source is unavailable.

Panel C, reports the United States sign reversal and the persistent predominance of adjustment for third countries under the reverse order.

G.4 Calibration and climate projections

Table G5 collects the broader specification checks. Government-specific weights are clipped to $[0, 1]$, with the pooled weight assigned where identifying observations are insufficient. Elasticity changes re-estimate the common welfare weight. Agricultural-source alternatives retain the manufacturing projections and the principal agricultural input where the alternative source is missing. These exercises assess the specified inputs within the model rather than uncertainty about future climate and development as a whole.